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Churn Prediction and Lifetime Value Analysis: A Data-Driven Client Retention Strategy for Telecom Companies.

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Abstract

In the ever-increasing competition in the telecommunication sector, customer churn poses a great challenge. This affects the revenue of the company concerned and harms its reputation. This project uses machine learning to forecast the probability of customer churn and assess the customer lifetime value to create a solid data-driven client retention strategy. Using customer demographic data, a predictive model is created to foretell the likelihood of clients leaving the telecom service provider. In addition, determining the client's lifetime value can also enable the estimation of the possible net revenue generated over the customer relationship. Recall, precision, accuracy, the F1 score and the AUC ROC score are the performance indicators used to evaluate the prediction model's efficacy. The model's insights and forecasts are presented in a simple and user-friendly dashboard to encourage data-driven decision-making. The project's findings offer telecom firms practical advice and insights to successfully lower churn rates, increasing customer lifetime value, general satisfaction, operational effectiveness and revenue. This research provides useful insights and equips stakeholders in the telecom industry to create focused client retention strategies by integrating statistical analysis utilizing machine learning approaches.

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I. INTRODUCTION

The telecommunications sector operates in an extremely challenging environment. Therefore, customer turnover or churn is a major issue in this sector. Customer churn is the phenomenon where customers decide to cease to use a telecom firm's services. Customer churn has negative effects on the company's brand and customer loyalty in addition to the obvious financial consequences (Hawkins, D. L, et al., 2019). As a result, telecom businesses need to establish a solid data-driven client retention strategy.

A plan that forecasts the loss of customers and assesses the lifetime value of customers can be put into place using data analytics and machine learning approaches, this can offer useful insights for making well-informed choices. As the mobile Internet era develops, customers have an increasing choice of application providers to choose from as their expectations for online services increase. For communication organizations, it's critical to protect each customer's particular demands, especially those with high value. They also need to keep their clientele relatively stable. If the telecoms industry doesn't concentrate on satisfying customer needs and preserving client relationships, it runs the danger of losing some discontented consumers. When taking into account the following facts, telecom businesses have begun realizing how crucial it is to keep customers engaged in order to increase revenue:

1. The cost of acquiring a new telecom customer is six to ten times greater than that of keeping a current one. Additionally, gaining just one negative customer can ruin your prospects of obtaining eight to ten excellent ones.
2. Mobile operators have experienced loss of customer rates as high as 30%. A whole customer base can transfer from one rival to another in slightly over three years.
3. Mobile providers can take up to a year to reach break-even on a new subscriber.

Therefore, it is essential that companies in the telecom industry establish a data-driven customer retention strategy. In order to effectively retain clients, proactive actions are required due to the intense competition and continually changing customer preferences. Only an in-depth examination can help stakeholders make decisions. Following this recognition of the significance of customer satisfaction, telecommunication firms have been undertaking ambitious research projects designed to uncover the mysteries of customer satisfaction in the telecom industry and produce takeaways that can direct their decision-making procedures and future strategies (Saroha, R. et al., 2020).

Many studies have shown how effective machine learning technologies are in predicting this circumstance. This method is utilized by studying previous data. Using statistical analysis and machine learning, we can simplify this task for businesses to help them prioritize their resources. Telecom firms could weigh their retention efforts, manage resources effectively, and develop individualized plans to improve customer pleasure and optimize revenue generation by precisely estimating attrition and customer lifetime value. If done early, identifying customers most likely to depart the company could represent a significant extra revenue source.

Establishing a churn prediction model that uses both past and current data analysis to extract critical information to aid in decision-making and uncover hidden linkages and patterns that can be used to forecast the future is crucial for the business. Businesses can take proactive steps to provide excellent customer service and lower customer attrition. The focus of research in the telecommunications industry has shifted to the use of big data analysis technology for data mining, anticipating customer churn, and mitigating the occurrence of customer churn (Kayaalp, F, 2017).

It is possible to extract and combine elements from past customer behaviour data, including customer data, usage behaviour, consumption behaviour, online trajectory, and other important indicators, in order to precisely evaluate the status or trend of customer turnover. This enables companies to target customer retention, which is one of the most efficient ways to address customer churn and is crucial for the long-term development of the business.

Data pre-processing, exploratory data analysis, the deployment of suitable machine learning models, and the creation of an intuitive dashboard are all part of this research. This research will help stakeholders develop client retention tactics in the telecom industry by integrating machine learning technology and big data analysis, providing considerable advantages in terms of customer happiness, revenue growth, and operational efficiency (Mohamed, F. A., et al, 2023).

In summary, this study will enhance client retention approaches in the telecom industry by providing beneficial data to assist with decisions and promote long-term development.

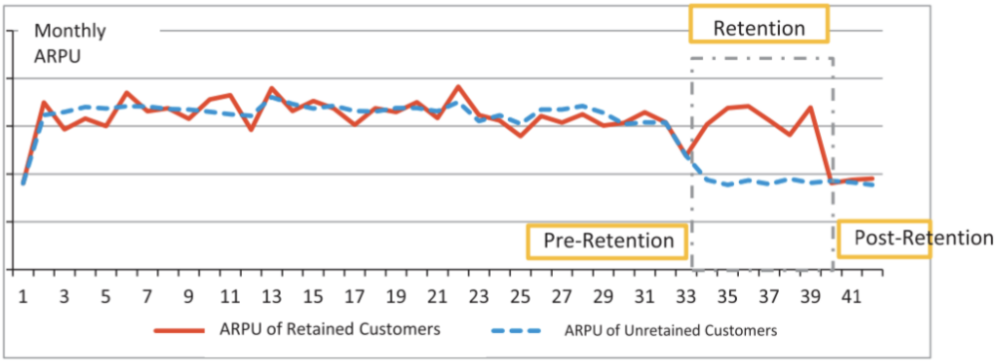


Figure 1 Impact of Customer Retention on Overall Revenue. (Peng, J. et al, 2013)

II. LITERATURE REVIEW

Peng, J. et al. (2013) demonstrate how average revenues per user (ARPU) prior to and following the retention period affect the entire revenue of a certain telecom organization. Additionally, the author uses retention strategies and monitors how they influence specific customers. The analysis results indicated that, although the customers were retained, the company's overall revenue declined after the retention period (Fig. 1). This can be the case since the revenue of the business is not significantly boosted by the retained customers. However, the investigation is unable to determine how to approach this. In order to deploy strategies early and retain more customers, the author goes on to recommend employing algorithms to forecast telecom churn.

Similarly, Shafei, I., et al. (2016), offered significant information for practitioners in the mobile service provider industry to focus their efforts on areas that can promote consumer loyalty and retention. The study offered insights into the aspects impacting customer satisfaction and loyalty in telecommunications company operations. To give a more comprehensive understanding of the elements influencing consumer loyalty and happiness, However, the study does not compare its findings with those in other markets or industries. It would also be beneficial to investigate the potential moderating impact of other factors on the link between customer happiness and customer loyalty, such as trust or perceived value. By utilizing machine learning or deep learning approaches, the study could have enhanced the study's outcomes.

With some further research, in order to identify churned consumers as well as understand the causes of customer turnover, Ullah, I., et al. (2019) suggested a churn prediction model for the telecom industry. This model makes use of a correlation attribute ranking filter and feature selection based on information gain. The models applied were J48, Naive Bayes, Multilayer Perceptron, Random Tree, Decision Stump, and Random Forest. For churn classification, Random Forest and J48 outperformed other algorithms, reaching an accuracy of 88.63% and 88.58% accuracy, respectively. When combined with ensemble algorithms like AdaboostM1 and Bagging, Decision Stump and Random Tree saw an increase in accuracy to 83.95% and 88.61%, respectively. Random Forest surpassed other algorithms in terms of accurate categorization. The model construction process took longer and had lower performance for models such as Naive Bayes, Multilayer Perceptrons, and Logistic Regression.

Furthermore, Li, W., & Zhou, C. (2020) also used the random forest model in order to test current data and forecast future churn based on past data in the telecom business. SVM, linear regression, random forests and logistic regression are just a few of the algorithms that could be employed for this, however, the random forest technique was chosen due to its benefits in terms of accuracy, robustness, and usability.

The results show that the random forest-based churn model did a good job of foretelling client churn. The random forest model also provided rules that explained the turnover patterns of particular customer groups by analyzing customer behaviour and finding essential attributes through segmentation. With a precision rate of about 80% and a recall rate of 40% during testing, the model's evaluation based on accuracy rate and coverage rate shows favourable outcomes.

Schaeffer, S. E., et al. (2020) forecasted client retention in a business-to-business environment utilizing machine learning techniques with data on prepaid unitary services. The idea was to spot clients who could be at risk of leaving before they stop being clients so that businesses can take proactive steps to ensure their retention. To train classification models, different time-series properties were retrieved from client transaction records. These properties included trend and level, magnitude, auto-correlations, and Fourier coefficients. With the maximum accuracy and specificity, the Random Forest model performs better than the alternatives. The linear Support Vector Machine (SVM), however, turns out to be a good and quicker choice for real-time forecasting on huge data sets. The study could further be strengthened by putting the machine learning models to the test on various datasets to determine if they perform well in other circumstances. Moreover, it would be beneficial to research the cause behind the models occasionally producing inaccurate predictions as well as various methods for choosing the most pertinent attributes for the models.

Similarly, the authors Bhuse, P., et al. (2020) focus on applying machine learning and deep learning approaches to anticipate customer attrition in the telecom business too. Models, such as Deep Neural Networks, Random Forest, SVM, XGBoost, Ridge classifier, and SVM were used. The testing data before grid search gave the highest accuracy to the Random Forest model, 90.96%. The models were optimized using grid search and cross-validation, which increased accuracy by up to 91.26%. The paper could, however, benefit from more thorough justifications for model selection, consideration of larger and more varied datasets, comparative analysis with other studies, exploration of the entire potential of deep learning approaches for churn prediction, as well as useful implications for telecom firms.

Furthermore, Jain, H. et al. (2020) have made use of two machine learning methods, Logit Boost and Logistic Regression. The study presents a churn prediction model. The results of the study show that the machine learning methods Logit Boost and Logistic Regression were equally effective at forecasting customer attrition in the telecom industry. Both models' accuracy percentages were about 85%, proving their potency in identifying future churners. A slightly higher Kappa number for the Logit Boost model indicated that it performed better in terms of prediction accuracy. Additionally, the study further compared other performance metrics for both strategies, including mean absolute error, relative absolute error and, root mean square error. In terms of overall performance on these measures, both models scored equally. The paper recommends investigating hybrid models and evaluating various approaches including rough set, SVM, and neural networks for better churn prediction. Likewise, Al-Mashraie, M. et al. (2020) offer insightful information about customer retention in the telecom business and the impact of churn prediction. The article focuses on predicting customer churn in the telecom industry in order to identify the key factors that influence customer retention. The study evaluates the effectiveness of several churn prediction models, such as decision trees, random forests, logistic regression, and support vector machines. The most accurate churn prediction model, when compared to the others, was the logistic regression model. Additionally, the push-pull-mooring (PPM) framework was used to comprehend how different features affect customer turnover behaviour from push, pull, and mooring viewpoints. One of the most significant elements influencing customer attrition is the drop-percentage push factor, demonstrating that customers are more sensitive to service quality in terms of call drops.

Similarly, Saheed, Y. K. et al., (2021) presented a novel feature selection strategy that combines Information Gain and Ranker methods. With feature selection, the model obtained an accuracy of 95.02% and an accuracy of 92.92% without feature selection, according to performance evaluation utilizing F-measure with 10-fold cross-validation, accuracy and, precision, and. In every evaluation metric, the IG+R-RF model performed better than other models, while IG+R-NB performed the worst. This study introduced an innovative feature selection approach that blends Information Gain and Ranker techniques. According to performance evaluation using F-measure with 10-fold cross-validation, accuracy and precision, the model's accuracy was 95.02% using feature selection and 92.92% without feature selection. The IG+R-RF model outperformed the competition in every assessment criterion, but the IG+R-NB model performed poorly. However, the study does not document great length on the constraints and difficulties that the research had to deal with. A discussion of the generalizability, scalability, and potential data biases of the suggested approach would have been advantageous.

Furthermore, the relevance of customer relationship management (CRM) and client retention in the telecom industry is covered in VLN, R. K., et al. (2021). It emphasizes how support vector machine (SVM) algorithms, in particular, play a key role in improving client retention methods. In order to comprehend consumer behaviour and the elements causing churn, the conceptual framework examines the marketing mix theory and expectation-confirmation theory. The research emphasizes the importance of churn prediction in the telecom industry and suggests using SVM for effective data analysis and churn prediction. While proposing the idea of churn prediction using the SVM algorithm, the research does not include any empirical support or real-life case studies.

To effectively anticipate churn, the suggested SVM-based approach must be tested and evaluated on real telecom datasets. Furthermore, The Decision Tree method was the most accurate of the four evaluated algorithms used in a study by Khalid, L. F. et al. (2021). Its distinctive structure, which recognizes the most essential features and creates a tree-like structure to partition the data depending on these qualities, is responsible for its excellent accuracy. This feature enables the Decision Tree to produce results more quickly and accurately than the other algorithms. Although the Decision Tree method performs the best in terms of accuracy, it is important to note that the other algorithms were not far behind. With an accuracy of 91%, the Random Forest algorithm came in second place. The accuracy of the Neural Network and Naive Bayes algorithms was 90% and 88%, respectively.

Another more recent investigation by Sharma, A. et al. (2022), suggested an algorithm to find the main reasons why customers leave and how to increase customer retention. The study concentrates on customer churn prediction in the telecom sector. For predicting customer churn, it analyses the effectiveness of several conventional data evaluation approaches, including Logistic Regression, XGBoost, Decision Tree, Random Forest, Support Vector Machine (SVM), Gradient Descent Boosting, Cat Boost and Light Gradient Boosting. The findings demonstrate that after reducing the dataset, XGBoost outperforms other methods and obtains the best accuracy (81%). The study's findings support improved customer retention efforts in the telecom sector because the proposed algorithm can properly forecast customers who are likely to leave. Likewise, in their study, Zadoo, A., et al. (2022) focus on employing machine learning methods for customer segmentation and churn prediction in the telecom industry. In the study, Churn prediction is a problem that has been solved using binary classification, and common techniques are employed for it, including random forests, decision trees, logistic regression and multi-layer perceptron neural networks. The article suggests a system that compares the effectiveness of various churn prediction models after training them on pre-prepared data. Model evaluation is done using metrics like recall, precision, AUC-ROC curve and the F-measure. According to the study's findings, the suggested approach can assist telecom businesses in identifying possibly churning clients and developing effective retention plans to lower customer churn. However, no mention is made of particular data or a critical evaluation of the performance and effects on customer retention tactics of the suggested system.

Furthermore, to predict customer turnover, Fujo, S. W. et al. (2022) recommended employing deep learning techniques like the Deep-BP-ANN model. This research was conducted using real-world datasets from Cell2Cell (51047 customers) and IBM Telco (7043 customers). It was found that the XG Boost method was more successful in identifying the features that caused the churn. The Deep-BP-ANN model, however, beat the other classifiers in terms of F1-score, AUC, recall, accuracy, and precision. On the IBM Telco dataset, the 10-fold CV approach performed best (accuracy 86.57%, AUC 86.57%, recall 94.45%, F1 -Score 87.55%, and precision 81.59%). Additionally, employing the 10-fold CV approach, the Cell2Cell dataset achieved higher results (accuracy of 73.90%, AUC of 73.90%, recall of 89.74%, F1 -Score of 77.47%, and precision of 68.15%) (Fujo, S. W. et al., 2022). The 10-fold CV findings and the straightforward holdout obtained by the Deep-BP-ANN model in the study are shown in Figure 2. Prabadevi, B. et al (2023) in their study have also made use of machine learning techniques to solve the issue of customer churn. The study evaluates several machine learning techniques, including K-Nearest Neighbours, Logistic Regression, Random Forest, and Stochastic Gradient Booster. These algorithms' respective accuracy ratings are 83.9%, 82.6%, 82.9%, and 78.1%. The Stochastic Gradient Booster (SGB) algorithm, which received the highest accuracy ratings during the study, performs the best. The authors use Grid Search CV and Randomized Search CV for hyperparameter tuning. ROC, AUC, precision, recall, and accuracy are some of the metrics that are used to assess the model's efficacy. The study finds that SGB surpasses the other models in terms of accuracy, making it the most appropriate algorithm for customer churn prediction.

Finally, Taskin, N. (2023) in their study address the crucial problem of customer churn in the telecom industry and the value of client retention for businesses. It emphasizes how important machine learning-based churn prediction models are for reducing customer attrition and boosting customer loyalty. The study's proposed model analyses large amounts of data and makes use of several machine learning techniques to predict customer churn, including Adaboost, Gradient Boosting, Light Gradient Boosting Machine (LGBM), and Extreme Gradient Boosting (XGBoost), Logistic Regression, Support Vector Machine, and Random Forest. With XGBoost and LGBM classifiers reaching the best accuracy score of 95.74%, the trial results show the usefulness of the suggested model. It is seen that ensemble classifiers (XGBoost, Gradient Boosting, LGBM, and RF) outperform linear models (LR, SVM, and KNN) on the dataset, perhaps as a result of the dataset's non-linearity and high dimensionality. In order to give a thorough review, the study also assesses the model using a variety of reliable evaluation criteria, including precision, recall, specificity, G-Mean, Roc-Score, and MCC. In each of these criteria, the XGBoost classifier regularly beats competing models, demonstrating its potency in forecasting customer attrition.

	Deep-BP-ANN results obtained from Holdout and 10-fold CV (IBM Telco)		*Deep-BP-ANN results obtained from Holdout and 10-fold CV* (Cell2cell)	
	Simple Holdout	**10-fold CV**	**Simple Holdout**	**10-fold CV**
Accuracy	88.12%	86.57%	79.38%	73.90%
Precision	84.71%	81.59%	74.50%	68.15%
Recall	93.05%	94.45%	89.32%	89.74%
F1-Score	88.68%	87.55%	81.24%	77.47%
AUC	88.11%	86.57%	79.38%	73.90%

Figure 2 Model Performance Comparison: Deep -BP-ANN (Fujo, S. W. et al, 2022)

The third phase of this research will be the focus of discussion that is the Customer Lifetime Value Analysis. After utilising deep learning techniques as suggested by Fujo, S. W. et al. (2022) to predict customer turnover, we may analyze the customers to determine the customer lifetime value. This will help to address the issue stated by Peng, J. et al. (2013). Stakeholders in the business can identify the customers who generate the bulk of its total revenue and put retention strategies in place to keep them around even after the retention period ends. Consequently, the company's revenue will continue to be more stable even after the retention period concludes.

III. CRITICAL ANALYSIS:

The techniques, results, and implications of the studies described in the literature review are critically examined in this section. The goal is to showcase the state of customer churn prediction and retention tactics by looking at the strengths, flaws, and trends across the research.

The studies described above have addressed a wide range of strategies for predicting customer churn, including statistical analysis, machine learning algorithms, and deep learning methods. Notably, the studies have explored both traditional models like Logistic Regression and sophisticated ensemble models like Random Forest and XGBoost. While precise churn prediction was the main goal, numerous studies additionally extended their study in order to identify the major factors affecting churn.

- **Methodological Evaluation:**
  From a methodology perspective, the studies have shown varying degrees of sophistication showcasing a wide spectrum of approaches to churn prediction. For example, Peng, J. et al. (2013) explained the complexities of average revenue per user (ARPU) before and after retention periods. He shed light on how retained customers might not necessarily lead to an increase in the overall revenue of the company. However, the study fails to provide concrete strategies to address this challenge. On the other hand, Ullah, I., et al. (2019) offer a churn prediction model that uses correlation attribute ranking various algorithms. Their comparison of different models like Random Forest and J48 shows the potential of ensemble models in achieving a higher accuracy as compared to the other models. Similarly, Li, W. & Zhou, C. (2020) employed the random forest model to forecast the customer churn based on historical data, this leveraged its accuracy, robustness and usability.
- **Robustness and Applicability:**
  The robustness and applicability of the models have consistently come up as one of the major issues across the investigations. In order to assure real-world applicability, multiple studies have recognized the necessity to validate their models on larger and more varied datasets. Furthermore, the scalability of the models, which is crucial for implementing solutions on a large scale in real-world operations, has not been adequately addressed in the majority of studies. For instance, Fujo, S. W. et al. (2022) introduce the Deep-BP-ANN model and show its superior performance. However, here too the scalability of this approach on larger datasets. On the other hand, Saheed, Y. K. et al., (2021) stresses the need to test their innovative feature selection approach on diverse datasets to ensure its effectiveness.
- **Insight and Practical Implications:**
  Considering the concerns expressed by Peng, J. et al. (2013) on the decline in revenue following the retention period, this study offers an innovative approach to the problem by integrating machine learning predictions with the analysis of customer lifetime value. This method makes it easier to find high-revenue generators and supports appropriate, targeted retention efforts. The revenue losses following retention may be mitigated by this approach. This also aligns with the proposition made by Sharma, A. et al. (2022) to identify the primary reasons for customer turnover and implement effective retention strategies.
- **Unaddressed Gaps and Future Directions:**
  The lack of studies examining temporal patterns and time series analysis in churn prediction is a notable gap in the literature. Furthermore, none of the research has specifically addressed how to identify churn reasons. To improve the efficacy of the churn prediction models, further study could look into these areas.
- **Contribution of this Study:**
  By using an integrated strategy that makes use of prediction models and customer lifetime value analysis, this study expands on the findings from previous studies. This study contributes to the developing field of customer churn prediction and retention tactics in the telecom industry by demonstrating the efficiency of the techniques used in reducing the revenue decline post-retention.

IV. Data Gathering, Description and Pre Processing

1. Data Gathering

Any data analytics and machine learning project needs to start with acquiring and preparing the necessary data. The dataset was downloaded from mavenanalytics.io. Each of the 7,043 clients of a Californian telecom operator in Q2 2022 has a profile in the Customer Churn dataset. Each record is unique to a single subscriber and includes information on that person's subscription services, geographic location, and status for the quarter (stayed, joined, or churned).

```
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 38 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Customer ID                        7043 non-null   object
 1   Gender                             7043 non-null   object
 2   Age                                7043 non-null   int64
 3   Married                            7043 non-null   object
 4   Number of Dependents               7043 non-null   int64
 5   City                               7043 non-null   object
 6   Zip Code                           7043 non-null   int64
 7   Latitude                           7043 non-null   float64
 8   Longitude                          7043 non-null   float64
 9   Number of Referrals                7043 non-null   int64
 10  Tenure in Months                   7043 non-null   int64
 11  Offer                              7043 non-null   object
 12  Phone Service                      7043 non-null   object
 13  Avg Monthly Long Distance Charges  6361 non-null   float64
 14  Multiple Lines                     6361 non-null   object
 15  Internet Service                   7043 non-null   object
 16  Internet Type                      5517 non-null   object
 17  Avg Monthly GB Download            5517 non-null   float64
 18  Online Security                    5517 non-null   object
 19  Online Backup                      5517 non-null   object
 20  Device Protection Plan             5517 non-null   object
 21  Premium Tech Support               5517 non-null   object
 22  Streaming TV                       5517 non-null   object
 23  Streaming Movies                   5517 non-null   object
 24  Streaming Music                    5517 non-null   object
 25  Unlimited Data                     5517 non-null   object
 26  Contract                           7043 non-null   object
 27  Paperless Billing                  7043 non-null   object
 28  Payment Method                     7043 non-null   object
 29  Monthly Charge                     7043 non-null   float64
 30  Total Charges                      7043 non-null   float64
 31  Total Refunds                      7043 non-null   float64
 32  Total Extra Data Charges           7043 non-null   int64
 33  Total Long Distance Charges        7043 non-null   float64
 34  Total Revenue                      7043 non-null   float64
 35  Customer Status                    7043 non-null   object
 36  Churn Category                     1869 non-null   object
 37  Churn Reason                       1869 non-null   object
dtypes: float64(9), int64(6), object(23)
```

Figure 3 Data Set Acquisition and Preparation.

2. Data Description

The dataset has 38 columns, each of which represents an individual customer attribute. The columns and corresponding datatypes are listed below.

1. Customer ID: Unique identifier for each customer (object).
2. Gender: Gender of the customer (object).
3. Age: Age of the customer (integer).
4. Married: Marital status of the customer (object).
5. Number of Dependents: Number of dependents of the customer (integer).
6. City: City of residence of the customer (object).
7. Zip Code: Zip code of the customer’s location (integer).
8. Latitude: Latitude of the customer's location (integer).
9. Longitude: Longitude of the customer’s location (integer).
10. Number of Referrals: Number of Customer referrals (integer).
11. Tenure in Months: Number of months the customer has been with the telecom provider (integer).
12. Offer: Offer type provided to the customer (object).
13. Phone Service: Whether the customer has phone service (object).

14. Avg Monthly Long-Distance Charges: Average monthly long-distance charges (float, with missing values)
15. Multiple Lines: Whether the customer has multiple phone lines (object, with missing values).
16. Internet Service: Type of internet service provided to the customer (object).
17. Internet Type: Type of internet connection (object, with missing values).
18. Avg Monthly GB Download: Average monthly gigabytes download (float, with missing values).
19. Online Security: Whether the customer has an online security service (object, with missing values).
20. Online Backup: Whether the customer has an online backup service (object, with missing values).
21. Device Protection Plan: Whether the customer has a device protection plan (object, with missing values).
22. Premium Tech Support: Whether the customer has premium tech support service (object, with missing values).
23. Streaming TV: Whether the customer has a streaming TV service (object, with missing values).
24. Streaming Movies: Whether the customer has a streaming movies service (object, with missing values).
25. Streaming Music: Whether the customer has a streaming music service (object, with missing values).
26. Unlimited Data: Whether the customer has unlimited data service (object, with missing values).
27. Contract: Type of contract with the telecom provider (object).
28. Paperless Billing: Whether the customer has paperless billing (object).
29. Payment Method: Payment method used by the customer (object).
30. Monthly Charge: Monthly charge amount (float).
31. Total Charges: Total charges incurred by the customer (float).
32. Total Refunds: Total refunds issued to the customer (float).
33. Total Extra Data Charges: Total extra data charges incurred by the customer (integer).
34. Total Long Data Charges: Total long-distance charges incurred by the customer (float).
35. Total Revenue: Total revenue generated from the customer (float).
36. Customer Status: Status of the customer (object).
37. Churn Category: Category indicating customer churn (object, with many missing values).
38. Churn Reason: Reason for customer churn (object, with many missing values).

	Customer ID	Gender	Age	Married	Number of Dependents	City	Zip Code	Latitude	Longitude	Number of Referrals	Tenure in Months	Offer	Phone Service	Avg Monthly Long Distance Charges	Multiple Lines	Internet Service
0	0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	9	None	Yes	42.39	No	Yes
1	0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	9	None	Yes	10.69	Yes	Yes
2	0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	4	Offer E	Yes	33.65	No	Yes
3	0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	13	Offer D	Yes	27.82	No	Yes
4	0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	3	None	Yes	7.38	No	Yes
5	0013-MHZWF	Female	23	No	3	Midpines	95345	37.581496	-119.972762	0	9	Offer E	Yes	16.77	No	Yes
6	0013-SMEOE	Female	67	Yes	0	Lompoc	93437	34.757477	-120.550507	1	71	Offer A	Yes	9.96	No	Yes

Figure 4 Data Description.

Internet Type	Avg Monthly GB Download	Online Security	Online Backup	Device Protection Plan	Premium Tech Support	Streaming TV	Streaming Movies	Streaming Music	Unlimited Data	Contract	Paperless Billing	Payment Method	Monthly Charge	Total Charges
Cable	16.0	No	Yes	No	Yes	Yes	No	No	Yes	One Year	Yes	Credit Card	65.6	593.30
Cable	10.0	No	No	No	No	No	Yes	Yes	No	Month-to-Month	No	Credit Card	-4.0	542.40
Fiber Optic	30.0	No	No	Yes	No	No	No	No	Yes	Month-to-Month	Yes	Bank Withdrawal	73.9	280.85
Fiber Optic	4.0	No	Yes	Yes	No	Yes	Yes	No	Yes	Month-to-Month	Yes	Bank Withdrawal	98.0	1237.85
Fiber Optic	11.0	No	No	No	Yes	Yes	No	No	Yes	Month-to-Month	Yes	Credit Card	83.9	267.40
Cable	73.0	No	No	No	Yes	Yes	Yes	Yes	Yes	Month-to-Month	Yes	Credit Card	69.4	571.45
Fiber Optic	14.0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Two Year	Yes	Bank Withdrawal	109.7	7904.25

Figure 5 Data Description (Continued).

Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	Customer Status
0.00	0	381.51	974.81	Stayed
38.33	10	96.21	610.28	Stayed
0.00	0	134.60	415.45	Churned
0.00	0	361.66	1599.51	Churned
0.00	0	22.14	289.54	Churned
0.00	0	150.93	722.38	Stayed
0.00	0	707.16	8611.41	Stayed

Figure 6 Data Description (Continued)

3. Data Pre Processing

In machine learning, the preparation of the data is crucial. To make accurate predictions and get novel insights, it is of the utmost importance that the data quality, method of handling missing or null values, and data transformation be of high quality. The data is suitable for further training and testing machine learning models after being cleaned and processed. Data preprocessing is an iterative process that involves experimentation and refinement. This makes it possible to precisely recognize biases or additional issues with the data, which eventually results in more robust and credible machine learning models.

i) Dropping Churn Category and Churn Reason:

During the data preprocessing stage, the "Churn Category" and "Churn Reason" columns were eliminated since they included a substantial number of missing values. Handling the missing values in the data is crucial since they can affect the model's overall performance. These two columns were unsuitable for inclusion in this analysis due to the large number of missing values in their situation.

ii) Handling missing values:

In the other columns of the dataset, there were also a few values that were missing. In order to assure reliable analysis, such a problem is required to be fixed. With the goal to fill in the missing information for numeric columns, the mean of each column was computed. For categorical columns, the mode was employed to complete the data gaps. Without affecting the model's performance, this method helped preserve the dataset's integrity and information.

```
Customer ID                          0
Gender                               0
Age                                  0
Married                              0
Number of Dependents                 0
City                                 0
Zip Code                             0
Latitude                             0
Longitude                            0
Number of Referrals                  0
Tenure in Months                     0
Offer                                0
Phone Service                        0
Avg Monthly Long Distance Charges  682
Multiple Lines                     682
Internet Service                     0
Internet Type                     1526
Avg Monthly GB Download           1526
Online Security                   1526
Online Backup                     1526
Device Protection Plan            1526
Premium Tech Support              1526
Streaming TV                      1526
Streaming Movies                  1526
Streaming Music                   1526
Unlimited Data                    1526
Contract                             0
Paperless Billing                    0
Payment Method                       0
Monthly Charge                       0
Total Charges                        0
Total Refunds                        0
Total Extra Data Charges             0
Total Long Distance Charges          0
Total Revenue                        0
Customer Status                      0
Churn Category                    5174
Churn Reason                      5174
```

Figure 7 Count of Null Values within the Dataset.

iii) Data type conversion:

Given that machine learning algorithms need numeric data to work, columns with the data type "object" had their numerical columns converted to "integer" data types, and their categorical columns converted to "categorical" data types. This made the data suitable with several machine learning algorithms for the prediction of customer churn. These data type transformations made sure that predictions were precise and reliable.

```
Customer ID                          object
Gender                             category
Age                                   int64
Married                            category
Number of Dependents                  int64
City                               category
Zip Code                              int64
Latitude                            float64
Longitude                           float64
Number of Referrals                   int64
Tenure in Months                      int64
Offer                              category
Phone Service                      category
Avg Monthly Long Distance Charges   float64
Multiple Lines                     category
Internet Service                   category
Internet Type                      category
Avg Monthly GB Download             float64
Online Security                    category
Online Backup                      category
Device Protection Plan             category
Premium Tech Support               category
Streaming TV                       category
Streaming Movies                   category
Streaming Music                    category
Unlimited Data                     category
Contract                           category
Paperless Billing                  category
Payment Method                     category
Monthly Charge                      float64
Total Charges                       float64
Total Refunds                       float64
Total Extra Data Charges              int64
Total Long Distance Charges         float64
Total Revenue                       float64
Customer Status                    category
```

Figure 8 Data Types of the variables in the Dataset.

iv) Handling Categorical data and Feature Scaling:

The handling of categorical data and feature scaling are essential next steps after dealing with missing values and changing datatypes. The dataset included a number of different categorical values, including "Offer, Phone Service, Multiple Lines, Internet Service, Internet Type, Online Security, Online Backup, Device Protection Plan, Premium Tech Support, Streaming TV, Streaming Movies, Streaming Music, Unlimited Data, Contract, Paperless Billing, Payment Method, Customer Status" Numerical values are necessary for machine learning algorithms to function. Therefore, it was crucial to represent this data numerically. The Label Encoder was utilized for this. The Label Encoder makes sure that every distinct categorical value has been transformed into its corresponding numerical value. This makes it easier for algorithms to effectively process and interpret categorical data while developing prediction models.

Similar to this, feature scaling standardizes the numerical data to make them uniform in scale and eliminate bias during the prediction process. This prevents any one feature from dominating the learning process given its larger scale. The Standard Scaler was employed to do this. The Standard Scaler was fitted to the data, the numeric features such as "Tenure in Months, Average Monthly Long-Distance Charges, Average Monthly GB Download, Monthly Charge, Total Charges, Total Refunds, Total Extra Data Charges, Total Long Distance Charges, Total Revenue" were extracted, and the data was then transformed into a new dataset.

	Tenure in Months	Avg Monthly Long Distance Charges	Avg Monthly GB Download	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	Offer	Phone Service	Multiple Lines	Internet Service	Internet Type	Online Security
0	-0.952994	1.257532	-0.587869	0.064221	-0.744500	-0.248313	-0.273300	-0.434195	-0.718872	0	1	0	1	0	0
1	-0.952994	-1.091613	-0.934015	-2.166367	-0.766962	4.602325	0.125055	-0.771190	-0.846108	0	1	1	1	0	0
2	-1.156740	0.609850	0.219805	0.330225	-0.882382	-0.248313	-0.273300	-0.725844	-0.914111	5	1	0	1	2	0
3	-0.789997	0.177815	-1.280161	1.102599	-0.460063	-0.248313	-0.273300	-0.457641	-0.500827	4	1	0	1	2	0
4	-1.197489	-1.336902	-0.876324	0.650712	-0.888318	-0.248313	-0.273300	-0.858681	-0.958059	0	1	0	1	2	0
5	-0.952994	-0.641051	2.700518	0.186006	-0.754142	-0.248313	-0.273300	-0.706555	-0.806980	5	1	0	1	0	0
6	1.573461	-1.145710	-0.703251	1.477568	2.481783	-0.248313	-0.273300	-0.049538	1.946607	1	1	0	1	2	1

Figure 9 Dataset after Feature Scaling.

Online Backup	Device Protection Plan	Premium Tech Support	Streaming TV	Streaming Movies	Streaming Music	Unlimited Data	Contract	Paperless Billing	Payment Method	Customer Status	Customer ID
1	0	1	1	0	0	1	1	1	1	1	0002-ORFBO
0	0	0	0	1	1	0	0	0	1	1	0003-MKNFE
0	1	0	0	0	0	1	0	1	0	0	0004-TLHLJ
1	1	0	1	1	0	1	0	1	0	0	0011-IGKFF
0	0	1	1	0	0	1	0	1	1	0	0013-EXCHZ
0	0	1	1	1	1	1	0	1	1	1	0013-MHZWF
1	1	1	1	1	1	1	2	1	0	1	0013-SMEOE

Figure 10 Dataset after Feature Scaling (Continued).

v) Resampling for class imbalance:

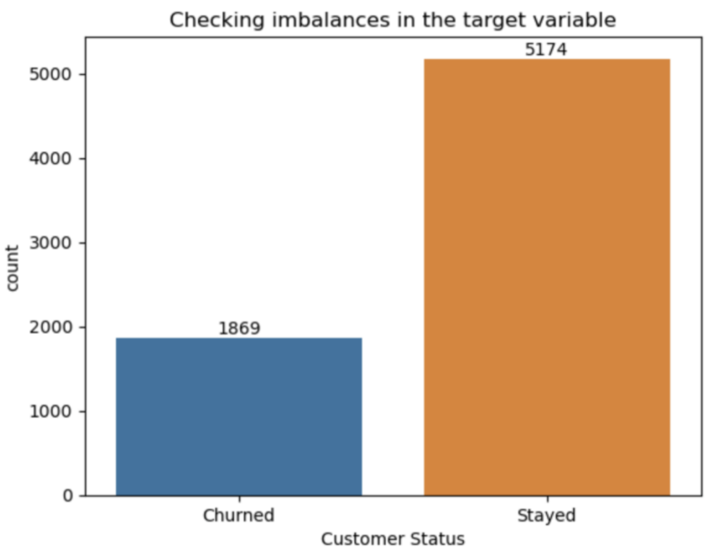


Figure 11 Bias in 'Customer Status' before resampling.

Another problem with the dataset was that the target variable, the "Customer Status" column exhibits class bias. Resampling was used as a countermeasure to this. Only 1869 instances of customers churned in the original dataset, compared to 5174 occurrences of customers who stayed. Due to the imbalance and the majority class's higher representation in the data, the model could favour the dominant class and produce biased predictions. The RandomOverSampler from the imbalanced-learn module was used to correct this class imbalance. In order to match the number of instances in the majority class (stayed customers), this technique generates synthetic samples in the minority class (churned customers). After resampling, the bias was lessened, with 7,043 instances of churned customers and 10,348 instances of stayed customers.

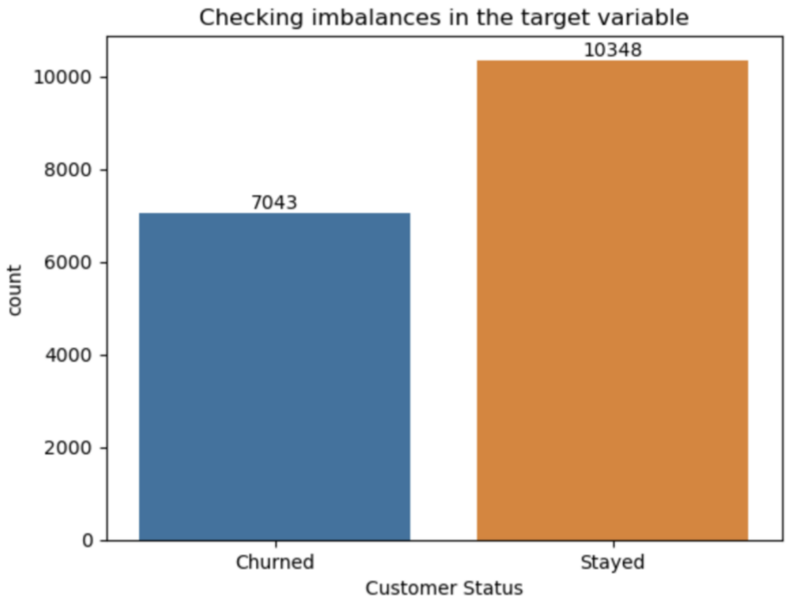


Figure 12 Bias in 'Customer Status' after resampling.

V. Exploratory Data Analysis

Exploratory Data Analysis is essential for the analysis of a dataset. It aids in thoroughly understanding the dataset and provides insightful information that can support data-driven decision-making. In order to uncover hidden patterns, correlations, and trends that would shed light on customer behaviour and its effect on turnover rates, a deep Exploratory Data Analysis on the customer churn dataset was done.

1. Correlation Analysis:

Before beginning the exploratory data analysis, we first looked at the relationships between the various variables and how those variables affect the Total revenue generated by the company. To accomplish this, we examined the correlation between the predictor variable and the target variable, or in this case “Total Revenue”, respectively.

To visualize this correlation matrix, a heatmap was used. This made it possible to understand the strength and direction of these relationships. The colours ranging from dark to light exhibit the strength of the correlation. Negative correlations are shown in light colours whereas positive correlations are shown in dark colours. The heatmap gives a brief overview of the key metrics, including “Tenure in Months”, “Number of Referrals”, “Total Charges”, “Total Long Distance Charges”, and “Monthly Charge”. All of these demonstrated a significant positive correlation with the total revenue.

“Tenure in Months” shows a strong positive correlation indicating that customers who are with the company longer contribute significantly to the total revenue of the company. Additionally, “Number of Referrals” also shows a strong positive correlation, which indicates that the customers who referred others to the company’s services also had a positive impact on revenue generation. Moreover, “Total Charges”, “Total Long Distance Charges”, and “Monthly Charge” also show a positive correlation indicating that customers who incurred a higher monthly or total charge due to services provided also contributed positively to the revenue generation.

This knowledge helps companies focus their attention on these key features when seeking to optimize their revenue generation strategies. By paying more attention to long-standing customers, incentivizing referrals and creating attractive packages and offering them to customers with higher charges, companies can increase their revenue generation.

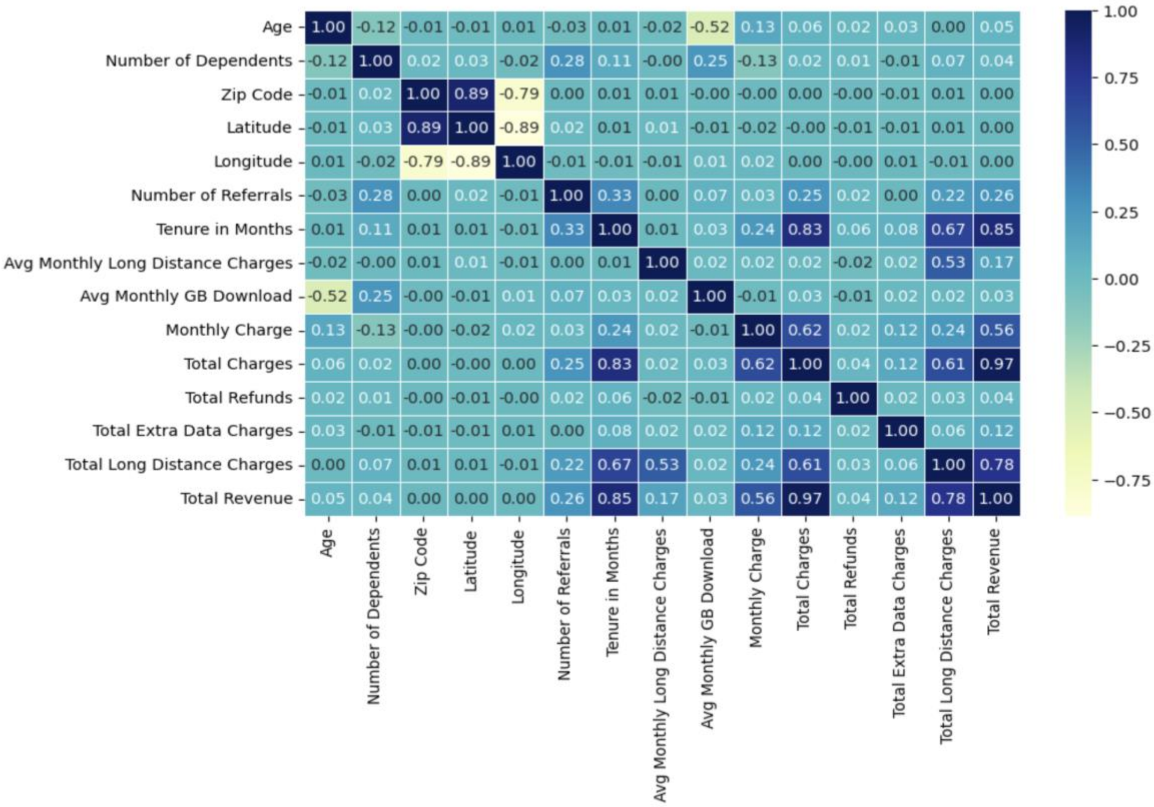


Figure 13 Correlation Analysis for Revenue Generation.

2. Feature Importance:

Understanding the significance of the different attributes with respect to customer churn prediction will help us obtain a better understanding of the characteristics and attributes in the dataset. By assigning the target variable “Customer Status” numeric values (1 for Stayed and 0 for Churned), the relationship between each attribute and the target variable was subsequently quantified. The bar graph below displays the absolute correlation for each feature, highlighting those that have the greatest influence on customer attrition.

The significance of “Tenure in Months”, suggests that customer loyalty plays a pivotal role in customer churn. Customers who have stayed with the company for a long time already will most likely not leave. Customers with shorter tenures are more likely to churn. Likewise, Customers who have a high “Number of Referrals” also displayed a high retention rate. Therefore, we conclude that the customers are more inclined to stick with an organization if they frequently refer others to its services.

By identifying influencing factors such as these, companies are able to develop meaningful strategies to retain valuable customers and consequently improve business performance. Therefore, In the end, utilising feature importance insights empowers businesses to make data-driven decisions, manage resources, and prevent customer churn, promoting long-term success and sustainability in today's competitive marketplace.

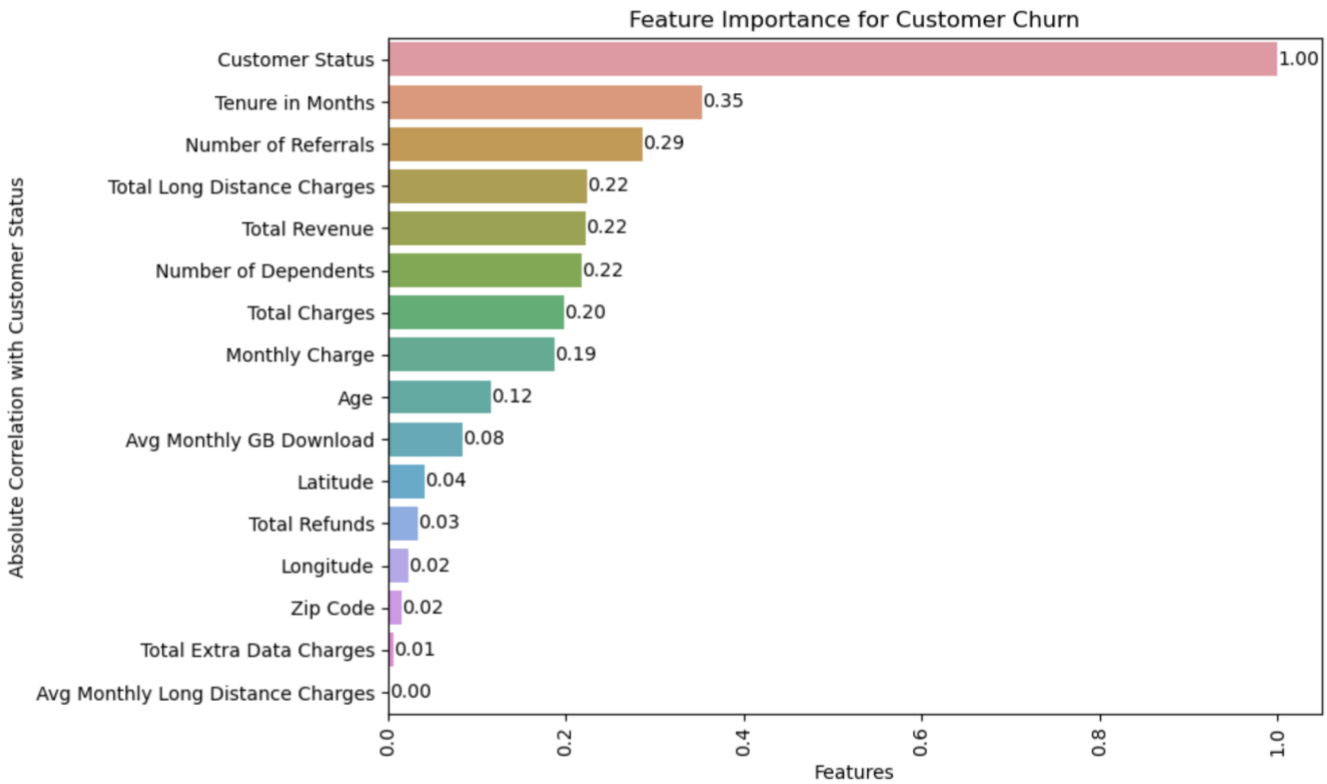


Figure 14 Feature Importance Insight in Customer Churn Prediction.

3. P – value for Chi-square Test of Categorical Features

The chi-square test was performed to determine and evaluate the relationship between each categorical variable in the dataset and the target variable, "Customer Status," which denotes whether a client has stayed with the company or churned. A higher p-value denotes a weaker or nonexistent relationship between the predictor and the target variable. With p-values of 0.3388 and 0.4866, respectively, two attributes, "Phone Service" and "Gender" show this sort of weak association. On the contrary, features with low p-values, especially those that are near zero, show a substantial correlation between specific categorical attributes and customer churn behaviours.

While characteristics that have high p-values have little influence on customer churn behaviour, these connections can act as critical predictors for precise machine learning models for estimating a loss of customers.

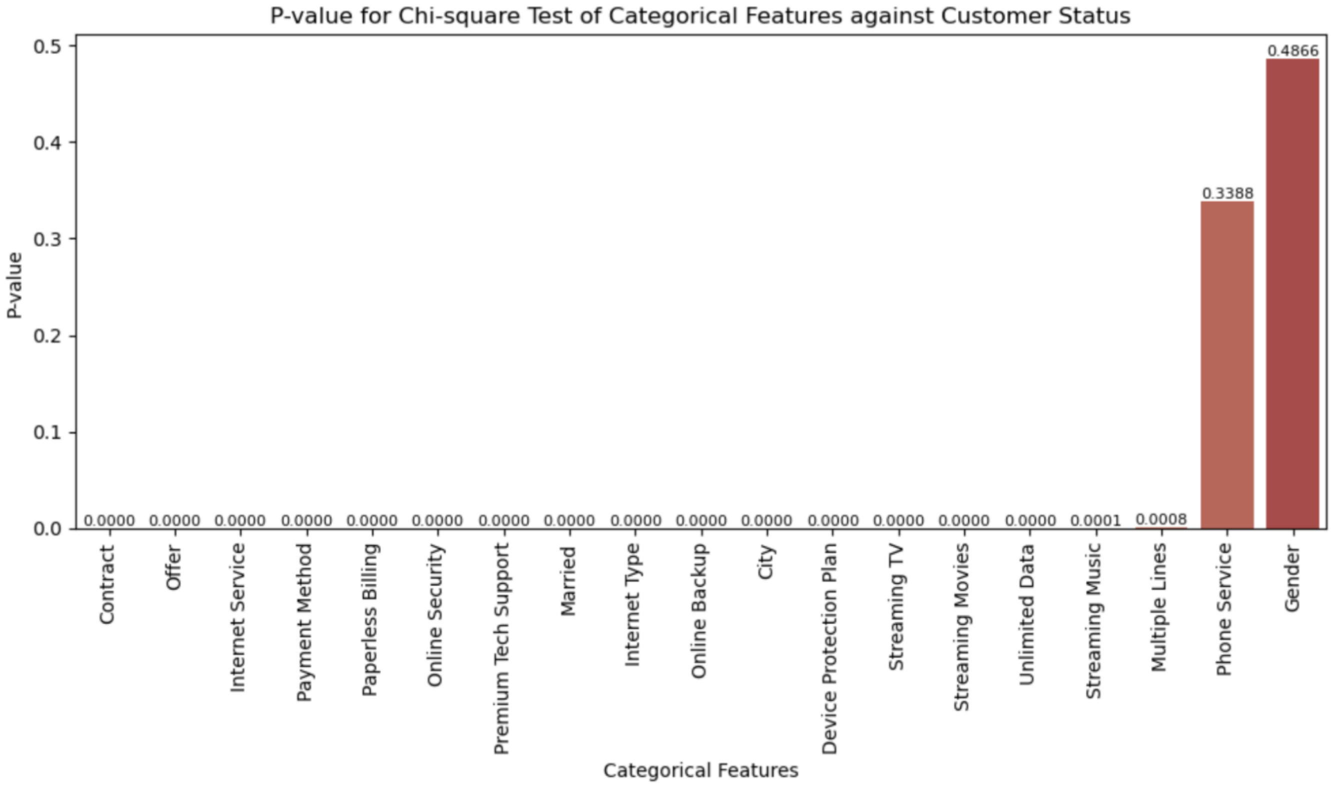


Figure 15 Chi-Square Test Results for Categorical Features: Influence on Customer Churn.

4. P – value for ANOVA Test of Continuous Features

The attribute "Customer Status" was used as the target variable in an ANOVA (Analysis of Variance) test to determine the relationship between each continuous attribute and the target variable. Similar to the Chi-square test, the p-values obtained are utilized to identify the relevance of these features in predicting customer churn.

According to the findings of the test, several continuous attributes in the dataset have higher p values. This indicates a weak relationship with customer churn. In comparison, "Longitude" (0.0435) and "Zip Code" (0.1717) had higher p - values. This demonstrates their lack of connection and the fact that they have no bearing on customer churn. Similarly, variables that have much higher p-values, like "Total Extra Data Charges" (0.5491) and "Avg Long Distance Charges" (0.8246) appear to have a weaker relationship with customer churn.

As a result, attributes with lower p-values should be taken into consideration due to the fact that they show higher connections with customer churn behaviour. These important continuous indicators, when included in predictive models, can help estimate customer turnover more accurately and direct organizations in putting targeted retention initiatives into place to lower it and improve overall business success.

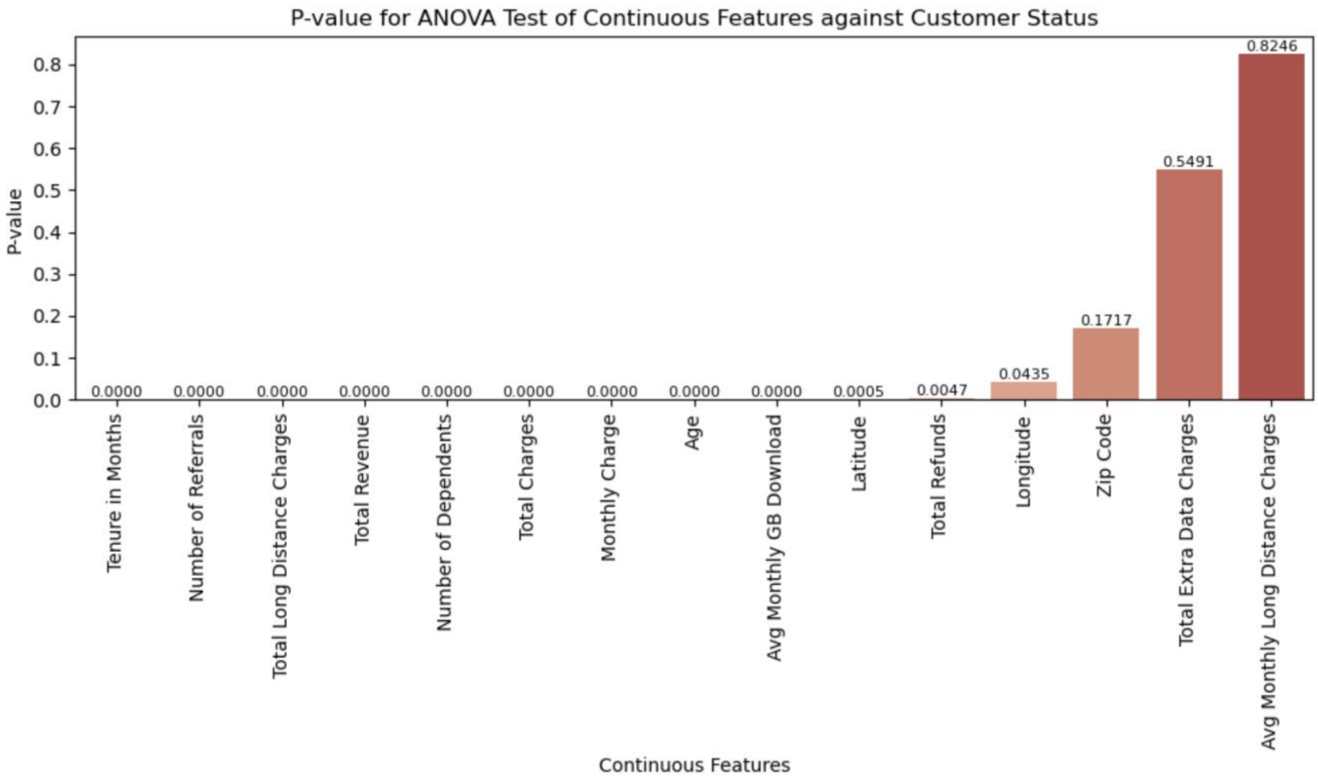


Figure 16 Analyzing Continuous Attributes with ANOVA: Relevance to Customer Churn.

5. Demographic Analysis:

Demographic data often yields an array of insights. Therefore, a distributional analysis of age and location was conducted. The age distribution of the company's customers can be seen in the histogram. This gives businesses the information they need to create customer-age-based marketing strategies that can target diverse age groups.

The top 7 cities with the greatest customer population are also shown on the count plot, which provides important data on the concentration of the company's services and possible regions for business expansion.

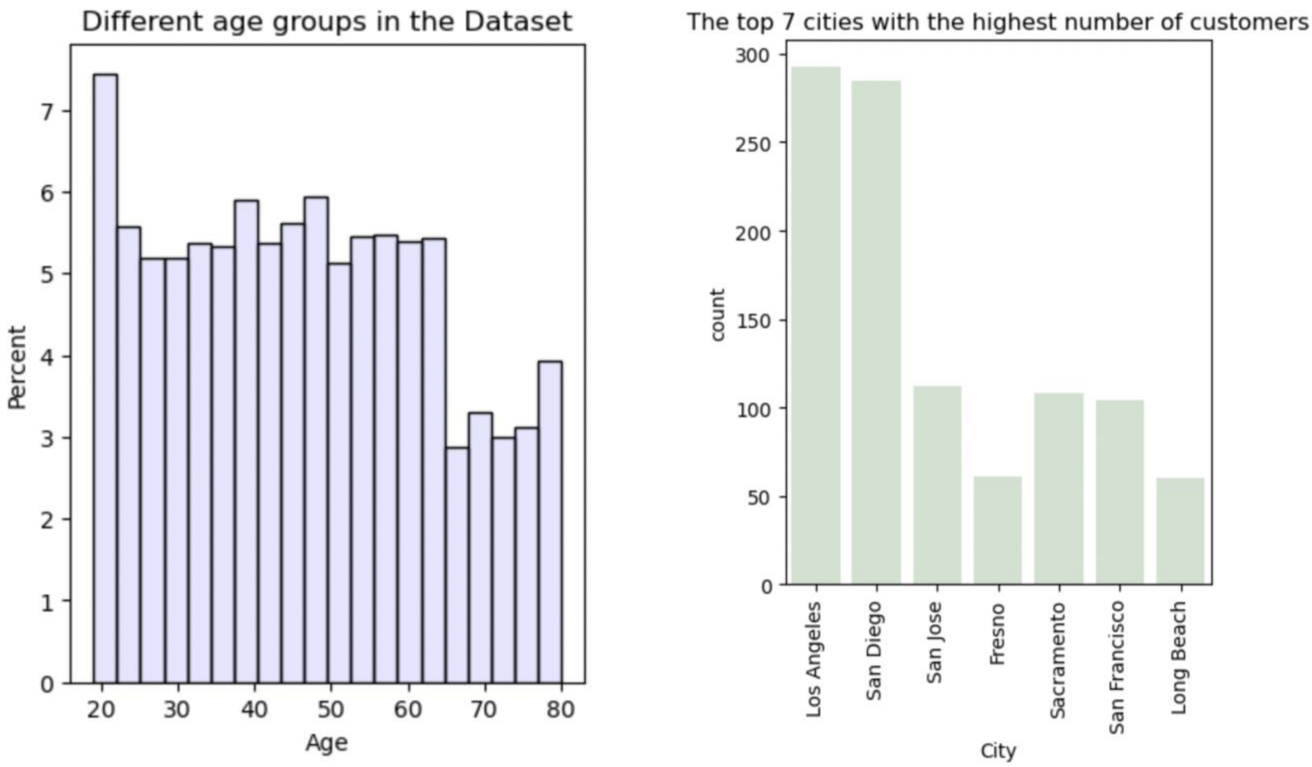


Figure 17 Exploring Customer Demographics: Age Distribution and Top Customer Locations.

4. Offer Distribution:

To maximize marketing efforts, it is essential to comprehend how the company promotes its offers. The pie graphic effectively illustrates this. The percentage of the deals that were more appreciated by customers can be observed in the pie chart below.

Notably, the chart shows more than half of the customers did not opt for any offer (55%) indicating that a significant portion of the customer base is unaccounted for any orders. Meanwhile, offer B has the highest percentage of people 12%, followed by "Offer E", "Offer D", "Offer A" and "Offer C" accounted for 12%, 11%, 9%, 7% and 6% respectively.

To increase customer engagement and retention, businesses can use this data to fine-tune their promotional campaigns and match their offerings with client preferences.

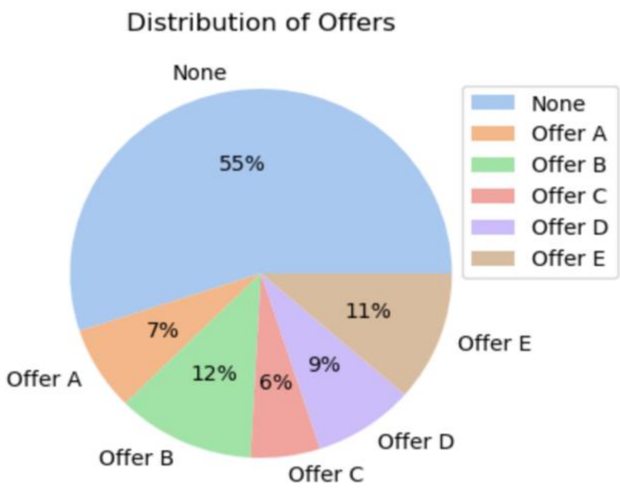


Figure 18 Customer Preferences for Company Offers.

5. Internet Service Preference:

Since customer happiness and churn rates are significantly influenced by the Internet service, we looked at what kinds of Internet users preferred. The frequency of each internet type that customers have selected can be easily identified by the count plot.

The companies offer DSL, Fiber Optic, and Cable internet services, according to the count plot. The most popular internet type among consumers is fibre optic, followed by cable, DSL, and subsequently.

As a result, businesses are better able to tailor their internet service offers to the needs and preferences of their current customers.

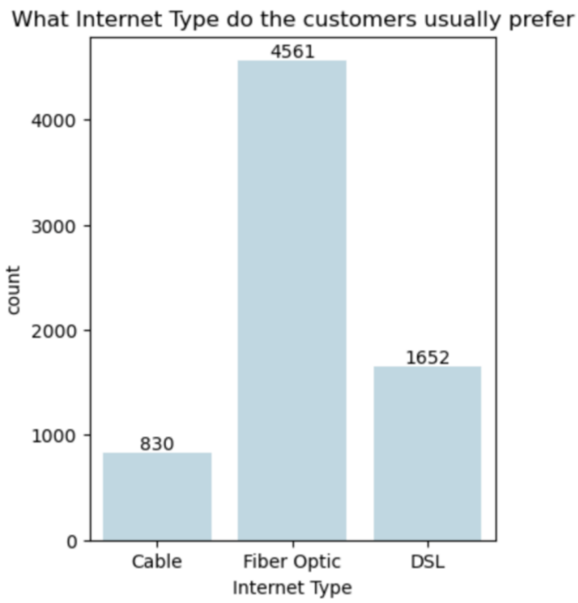


Figure 19 Internet Service Preferences among Customers.

6. Monthly Charges and Internet Type:

Understanding the connection between monthly charges and internet types is crucial. This aids in identifying potential pricing variances that may affect customer satisfaction and attrition rates. With respect to the customer's preferred type of internet connection, the box plot effectively depicts the average monthly cost.

Customers who prefer the fibre optic internet type here spend the most per month because the average monthly fee is around $80 to $81. This kind of internet connection is the most expensive. Additionally, customers who choose cable and DSL internet connections shell out, on average, $58 and $55 per month, far less than those who choose fibre-optic connections.

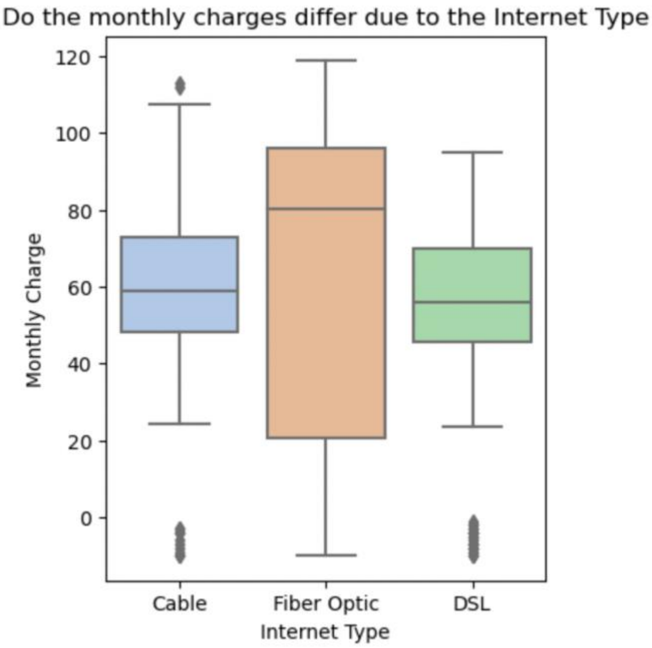


Figure 20 Monthly Charges vs. Internet Type.

7. Contract Type and Monthly Charges:

A bar graph was used to compare the median monthly charge that customers would pay for each contract type in order to study and assess the impact of the contract type on the monthly charges. With the use of this analysis, pricing decisions based on contract lengths and how they affect customers' spending can potentially be made.

The bar plot shows that the monthly recurring charges for a one-year contract and a month-to-month contract are roughly the same at $64.40 and $64.39, respectively. However, choosing a two-year contract enables users to take advantage of the same services for a considerably cheaper price of $61.41.

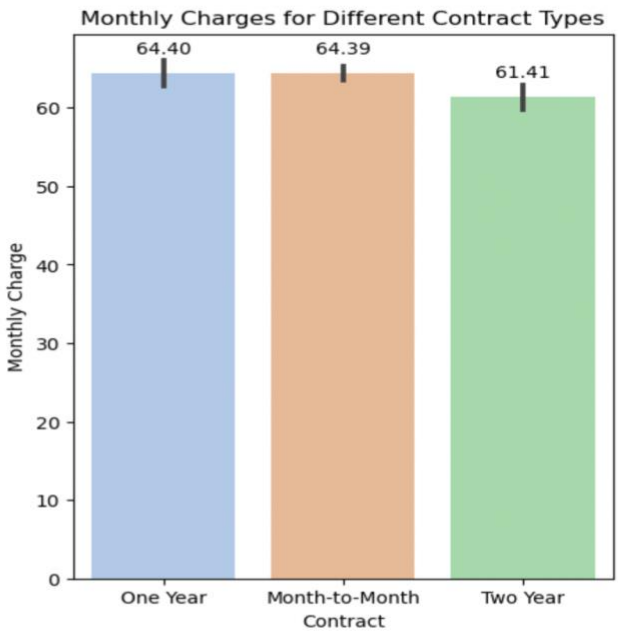


Figure 21 Monthly Charges for Different Contract Types.

8. Offer and Tenure in Months:

We acquire information and can evaluate the efficiency of various offers in keeping clients over time by comparing them against the tenure in months. The line scatterplot gives a general understanding of how a customer's preference for an offer type impacts how long they remained with the company or how long they stay engaged.

We can infer from the scatterplot that the clients who decided not to opt for offers have a varied range of retention times. There is little impact on their tenure from the absence of offers. On the contrary, consumers who selected Offer A have remained with the business the longest, indicating that this offer succeeded in retaining them. Customers who choose Offers E and D, nevertheless, abandon the company within 1 to 25 months.

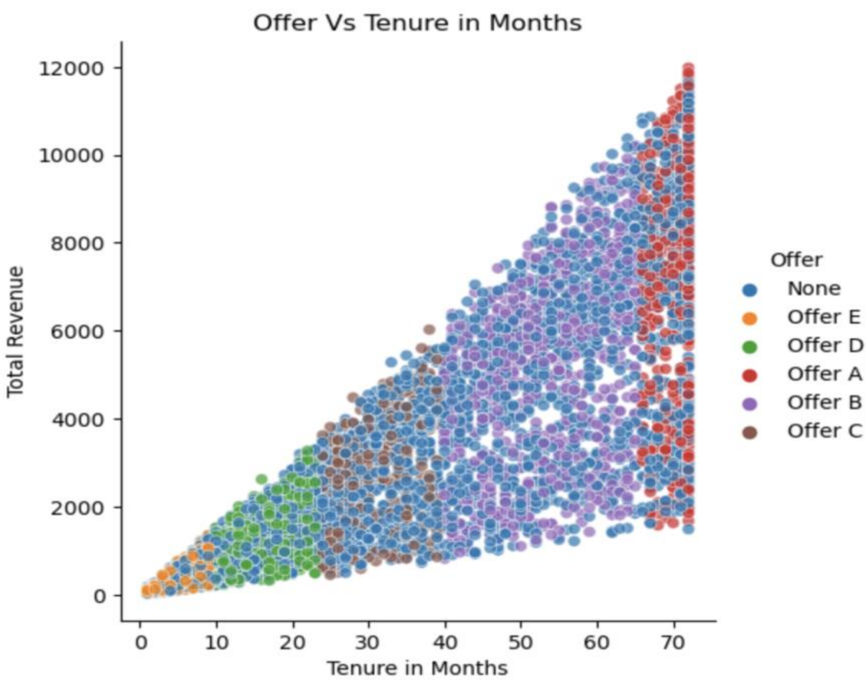


Figure 22 Offer vs. Tenure in Months.

9. Contract type and Tenure in Months:

A violin plot was used to effectively showcase the relationship between the contract types and how they affected the tenure of the customers in the company. The plot revealed interesting patterns and trends.
The violin plot indicates that a large number of people opted for the month-to-month contract at first, however, they showed a shorter tenure and most of them churned out within 10 months of joining. On the other hand, a smaller percentage of customers chose the two-year contract, but they displayed the highest level of loyalty. With mean tenure of about 62 months. Interestingly, the one-year contract had more customers than the two-year contract, yet, the mean retention rate of that contract type was about 41 months.
This gives us insights and helps companies to understand customer behaviours based on the contract durations.

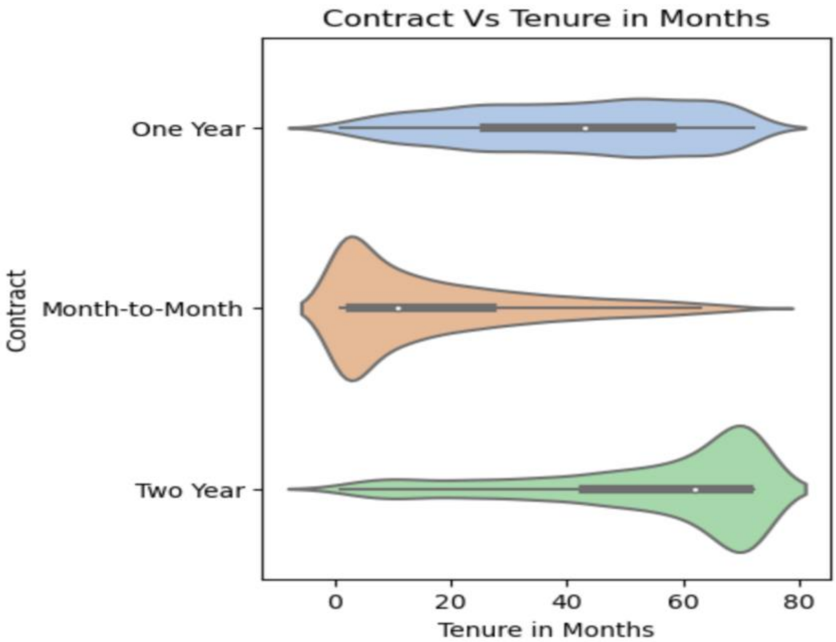


Figure 23 Contract vs. Tenure in Months.

10. Tenure in Months and Total Revenue

To find those customers with a greater amount of contribution, it is essential to analyze the relationship between tenure in months and overall revenue. The line plot clearly shows how the company's overall revenue varies over the tenure period, highlighting possible areas for revenue growth and retention tactics. The graph shows that as tenure increases, the company's overall revenue shows an upward trend. Notably, the total revenue reaches about $7,400 in just 70 months.
Thus, it can be established that long-term customers, or clients that stay with a company for a long time, significantly contribute to the overall revenue. Companies can employ targeted retention tactics to keep important customers and increase revenue potential by identifying these customers and their behavioural patterns.

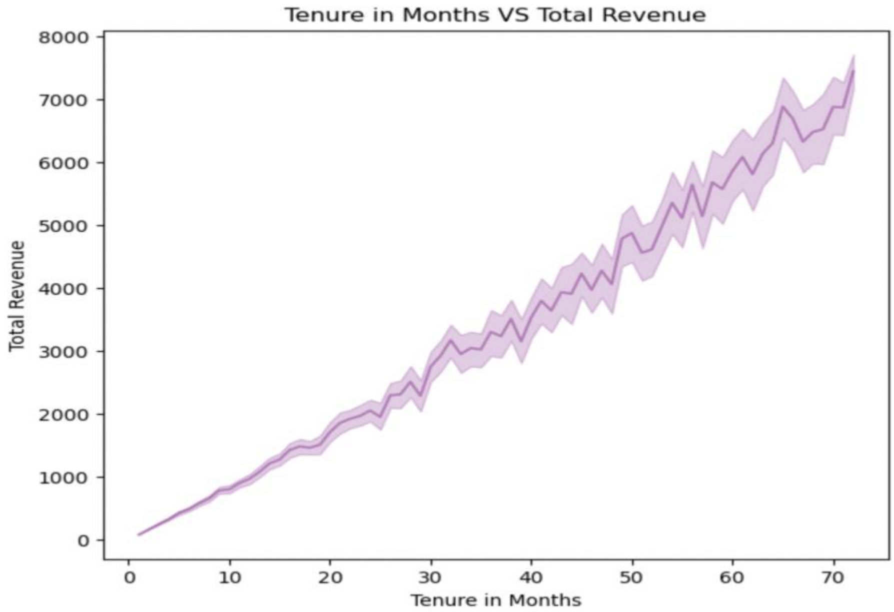


Figure 24 Relationship between Tenure in Months and Total Revenue.

11. Total Revenue and Monthly Charge across Offers and Contracts:

With the use of scatterplots, which depict the distribution of total revenue and recurring charges by using distinct colours to indicate offers and contracts, we were able to acquire a comprehensive knowledge of the dataset. This graph is designed to make it easier to identify the spending habits of the consumers based on numerous variables like the offers selected and the types of contracts.

The graph showed that the overall revenue tends to rise along with rising monthly fees. Additionally, the two-year contract category sees the most substantial gain in revenue from clients, followed by the one-year contract and then the month-to-month contract.

Customers who chose Offer A typically stay with a company for a longer period of time, which increases revenue overall. Additionally, among clients in the two-year category, Offer A is the most chosen option. It can also be noted that customers who choose a one-year contract tend to choose Offer B, whereas those who sign a month-to-month contract like Offer D and Offer E. Interestingly, customers who chose no offers had little to no impact on the churn rate. They were retained even though they didn't choose any of the offers, and some even boosted the revenue. Lastly, customers with two-year contracts spent less on average than those with one-year or month-to-month agreements.

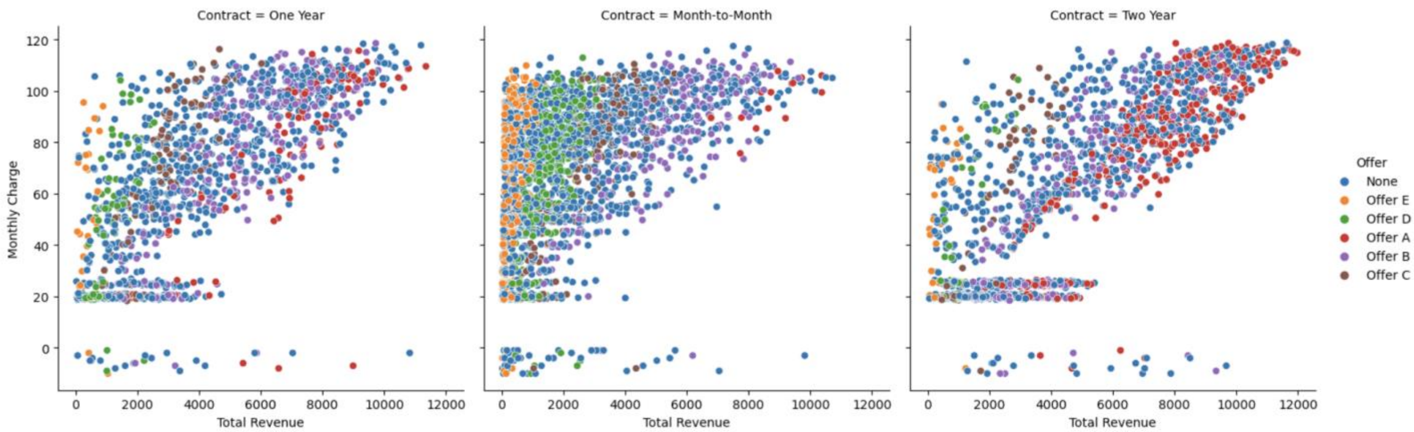


Figure 25 Total Revenue and Monthly Charge Distribution by Offers and Contracts.

Overall, the Telecom Customer Churn dataset exploratory data analysis has assisted in revealing key insights for well-informed decision-making. To do this, univariate, bivariate, and multivariate analyses have been used.
Some important variables as "Tenure in Months" and "Number of Referrals" have been recognized as having a significant impact on revenue. The feature importance analysis highlighted their predictive value in predicting customer churn likelihood.
The Chi-Square and ANOVA testing revealed correlations between certain continuous attributes and their influence on the chance of churn, whereas the Chi-Square tests helped identify the weak associations of attributes like "Phone Service" with churn.
The demographic analysis highlighted the preferences of the customer in terms of offers which was demonstrated with a pie chart and the internet service preferences. The analysis of contract types, tenure, and revenue demonstrated how specialized approaches could optimize revenue while satisfying customer demands.
Our EDA essentially provides companies with useful data that they may use to improve retention, adjust strategy, and manage a challenging market environment.

VI. METHODOLOGY

This section provides an overview of the methodology used to examine the telecom customer churn dataset. After pre-processing and analyzing the data, numerous machine learning models, including Support Vector (SVM), Random Forest (RF), Neural Network (NN), k-Nearest Neighbours (k-NN), Decision Tree, and Gradient Boosting, were used to assess the provided dataset. The following text provides a thorough explanation of how the models were used to train the data and to tune the hyperparameters.
After the Exploratory Data Analysis was concluded the data was scaled in order to ensure that all the features were on the same scale. The data was then divided initially into training (70%) and testing (30%) sets. A further division of the testing set into testing (50%) and validation sets (50%) was made to assess the model's performance. This ensured that all the models were evaluated on unseen data thus reducing the risk of overfitting.
To find the ideal parameters for each model, hyperparameter tuning using grid search was performed on all of the models using these training and testing sets. The models were adjusted to their optimum potential by selecting the parameter combination that produced the best performance metrics.
Below, we outline the models that were employed as well as the optimal parameters that were found using the grid search method for hyperparameter tuning.

1. Model Training

- **Support Vector Machine (SVM)**

Support Vector Machine (SVM) is a classification technique used for binary and multiple-class classification applications. When using SVM, the hyperplane that maximizes the margin between the classes in the feature space is identified. The margin is nothing more than the separation between the hyperplane and the closest data points for each class. Support vectors are the term used to describe these nearby data points. The Support Vector Machine method performs better in instances with complex decision boundaries and non-linear interactions. Mathematically, the SVM algorithm seeks to find the hyperplane represented by,

$$w^{\mathrm{T}}x + b = 0 \qquad (1)$$

Where the bias term is b and the weight vector is w.
To classify data points, apply the equation $w^{\mathrm{T}}x + b$. The hyperplane's points are divided into two classes: one for points on one side and another for points on the other. The SVM algorithm operates by maximizing the margin between these classes. The margin can be described as the perpendicular distance between the support vectors and the hyperplane. This is denoted mathematically as:

$$\text{minimize } \frac{1}{2} \| w \|^2 \quad \text{subject to } \; y_i \,(w^{\mathrm{T}} x_i + b) \geq 1 \text{ for all i} \tag{2}$$

Here, x_i represents the training data and y_i is the class label, and the inequality $y_i\,(w^{\mathrm{T}} x_i + b) \geq 1$, ensures that the data points are appropriately categorized within a specific margin.

- **Random Forest Model (RF):**

Random Forest is an ensemble learning technique which incorporates different decision trees to enhance prediction accuracy while lowering overfitting. The final prediction is based on the average score of the predictions from each decision tree, and each decision tree is trained on different subsets of the data.
In mathematics, the prediction of an individual tree is represented as,

$$\hat{y}_i = f(x_i) \tag{3}$$

Where, $\hat{y}_i$ is the predicted output for the i^{th} data point x_i, and f represents the decision trees prediction function. The Random Forest model prediction is obtained by aggregating the predictions from the individual trees.

$$\hat{y}\text{ RF} = \frac{1}{N_{trees}} \sum_{j=1}^{N_{trees}} \hat{y}_j \tag{4}$$

Where $\hat{y}$ RF is the Random Forest prediction followed by N_{trees}, which is the number of trees in the forest and $\hat{y}_j$ which is the prediction of the i^{th} decision tree.
The Random Forest model also provides a measure of feature importance which is based on the impurity reduction that is achieved by a feature when it is used for splitting nodes in the trees. The importance feature is represented by X_i. The average of the impurity reductions across all trees is used to calculate it.

$$Importance(X_i) = \frac{1}{N_{trees}} \sum_{j=1}^{N_{trees}} Impurity\ Reduction_j\,(X_i) \tag{5}$$

Where, $Importance(X_i)$ is the importance score of the features X_i, and $Impurity\ Reduction_j(X_i)$ is the reduction of the impurity in the i^{th} tree using feature X_i for splitting.

- **Neural Networks (NN):**

A machine learning technique called a neural network is one that was influenced by the design and operations of the human brain. The model is organized into layers through the association of nodes. There is an input layer, one or more hidden layers, and an output layer. The ability of neural networks to discover intricate patterns and connections within the data makes them effective for a variety of tasks, including regression and classification.
Mathematically, a Neural Network can be defined as a function f that maps an input x to an output $\hat{y}$,

$$\hat{y} = f(x) \tag{6}$$

The output layer $\hat{y}$ in the model is obtained through a sequence of consequent transformations across the different layers of the network. Each neuron in a particular layer receives input from neurons in the previous layer. An activation function is applied, the result of which is passed on to the neurons in the following layer.
The activation of a neuron can be mathematically represented as,

$$a_j = \sigma\left(\sum_{i=1}^{n} w_{ij}x_i + b_j\right) \tag{7}$$

Here, a_j stands for the activation neuron *j* in a layer, w_{ij} are the weights connecting the neurons in the previous layer to the neurons *j*, x_i is the input neuron and b_j is the bias term. σ signifies the activation function. Some of the activation functions include ReLU (Rectified Linear Unit), sigmoid function and Tanh (hyperbolic tangent).
Contrary to other machine learning models, feature importance scores are not automatically provided by neural networks. However, by observing and analysing the weights assigned to the input features in the hidden layers, feature relevance may be examined. For instance, characteristics with higher weights make a bigger difference in the final prediction made by the network. This indicates how crucial they were to the prediction.

- **K – Nearest Neighbours (k-NN):**

The classification method k-NN is easy to understand and straightforward. Based on the majority class of its k -nearest neighbours in the feature space, the k -NN model generates predictions. The model assumes that data points belonging to the same class are situated close to one another. K- NN chooses a data point's nearest neighbours and assumes that they are members of the same class. The distance between a new data point and all of the training points is calculated by k-NN before they can be classified. It then identifies the k nearest neighbours and assigns class labels to those neighbours based on how frequently they occur. Mathematically, the classification of a new data item *x* in a binary classification issue can be represented as,

$$\hat{y} = \arg\max_{y_i} \sum_{i=1}^{k} I_{(y_i)} \tag{8}$$

Where $\hat{y}$ is the predicted class label, y_i are the class labels of the k – nearest neighbours, and I is the indicator function. The distance between two data points x *and* x' can be calculated using Euclidean distance,

$$d(x, x') = \sqrt{\sum_{i=1}^{n} (x_i - x_i')^2} \tag{9}$$

Where n is the number of features and x_i and x_i' are the corresponding feature values of x *and* x'.
Like the neural network approach, the k-NN algorithm does not provide feature importance scores. Features with bigger scales may incorrectly influence the classification as k-NN bases its conclusions on the distances between the data points. In order to enhance predictions and guarantee that all characteristics contribute equally to the distance calculation, feature scaling is crucial in k-NN.

- **Decision Trees (DT):**

The decision tree model is a flexible classification approach that divides the feature space into subsets according to the various input values for features in the dataset. In this model, each division is carefully chosen to maximize the separation between each of the distinct values. As a result, the decision tree model is particularly effective for both binary and multi-class classification tasks.

The algorithm operates on the idea of making decisions by traversing the tree from the root to the leaf node. Here, each leaf node represents a class label, whereas each interior node reflects decisions based on the respective features.

A decision tree model can be conceptualised mathematically as a collection of if-else conditions. The following procedure is used by the decision tree to classify *x* into a class *C* given an input feature vector:

- Starting from the root node, traverse the tree based on the feature values.
- At each internal node, compare the values of a specific feature with a threshold value and follow the appropriate branch.
- Repeat until a leaf node is reached, which corresponds to the predicted class label *C*.

Decision trees automatically assign feature importance scores based on how effectively an attribute splits data. The extent to which a feature can minimize the impurity of the target classes in the data serves as a proxy for its significance. Gini impurity and entropy are two typical impurity measurements.

j represents the feature significance score for a feature. It is calculated by adding the normalised impurity reduction across all the internal nodes that make use of the feature *j*,

$$Feature\ Importance\ (j) = \sum_{internal\ nodes\ using\ feature\ j} \frac{samples\ in\ internal\ node}{total\ samples} \times impurity\ reduction \tag{10}$$

- **Gradient Boosting**

The Gradient Boosting model is an ensemble model that integrates the predictive power of several weak learners (which are frequently decision trees) to produce a strong predictive model. It advances stage by stage, creating an integrated model. Each new weak learner is trained in this model to fix the errors produced by previous ones. The overall goal of the Gradient Boosting model is to reduce the model's residual errors.

Mathematically, the prediction of Gradient Boosting model *F(x)* is the sum of the predictions made by multiple weak learners $h_i(x)$ with associated weights α_i

$$F(x) = \sum_{i=1}^{N} \alpha_i h_i(x) \tag{11}$$

Each weak learner $h_i(x)$ is trained to minimize the loss function with respect to the residual errors from the previous stage,

$$h_i(x) = \arg\min_h \sum_{j=1}^{m} L\left(y_j, F_{i-1}(x_j) + h(x_j)\right) \tag{12}$$

Where *L* is a loss function, y_j is the true label of a data point x_j, $F_{i-1}(x_j)$ *is the current model's prediction on* x_j and $h(x_j)$ is the prediction of the weak learner h_i on x_j.

Based on each feature's ability to contribute to the reduction in the loss function during the gradient boosting process, the Gradient Boosting model automatically assigns a value to each feature's significance. The average improvement in the loss function across all of the ensemble model's decision trees serves as an indicator of a particular feature's importance.

2. Hyperparameter tuning for the models:

- **SVM:**

In the Support Vector model, hyperparameter tuning entails determining the best values for parameters like the Regularization parameter C and the kernel selection. The trade-off between increasing the margin and reducing the classification error is controlled by the Regularization parameter C.
A low value of C results in a broader margin but a greater number of incorrectly classified points. Whereas a higher value of C minimizes misclassifications, however, it results in tighter margins. The most influential parameter combination that readily adjusts to data that is unseen can be determined using cross-validation procedures.

```
Best hyperparameters: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
```

- **Random Forest:**

There are a number of hyperparameters that affect how well the Random Forest Model performs. Some key parameters are listed below:

- N_{trees}: The number of trees in the forest
- max depth: The maximum depth of each tree.
- min samples split: The least number of samples necessary to split an internal node.
- min samples leaf: The least number of samples that must be present at a leaf node.

In a Random Forest Model, hyperparameter tuning entails determining the optimum combination of the mentioned parameters using methods like grid search or randomized search. To improve prediction accuracy and reduce overfitting, the Random Forest Model combines the diversity of several different decision trees. The individual decision tree forecasts are aggregated to form the mathematical foundation. The analysis of the impurity reduction caused by each feature's involvement when the tree node split further establishes the value of the feature. The goal of hyperparameter tuning is to find the ideal parameters to provide the best model performance.

```
Best hyperparameters: {'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 150}
```

- **Neural Networks:**

Numerous hyperparameters in the neural network model affect the final model's performance and training process. These are some of the hyperparameters:

- The number of hidden layers and the number of neurons in each layer
- The selection of activation methods to be applied to hidden layers.
- The learning rate, which determines the optimization step size.
- The batch size, or number of samples, refers to how many are used for each training update.
- The number of training epochs, or how many times the training dataset is passed through the network.

Neural networks mimic the behaviour of the human brain's networked neurons. They are mathematically based on how neurons are activated utilising weighted inputs and activation functions. Since feature importance is not provided by neural networks, the examination of these weights can help determine the significance of a feature.

```
Best hyperparameters: {'activation': 'relu', 'optimizer': 'adam'}
```

- **K- Nearest Neighbours:**

The number of neighbours to be employed for classification, or the principal hypermeter in the k-NN model, is k. While a bigger number might over smooth the decision boundary a lower value of k might cause noise influences. Cross-validation can be used to determine the ideal k value, which produces the best parameters for the model's performance on an unknown data set. The class label of a new data point is then predicted by the model based on the majority class of its closest neighbours. The key factor is the distance metric that is employed for evaluating similarity. As a result, hyperparameter tuning for the k-NN model aids in determining the ideal k value for improved generalisation.

```
Best hyperparameters: {'metric': 'euclidean', 'n_neighbors': 1, 'weights': 'uniform'}
```

- **Decision Trees:**

The generality and structure of decision trees may be impacted by a number of aspects. Some of the important factors include the maximum depth of the tree, the minimum samples required to split an internal node, and the minimum samples required at the leaf node.

The decision tree method, which builds hierarchical structures of decisions based on features, is a versatile classification technique that requires careful hyper-tuning to strike the right balance between complexity and generalisation. The model benefits from hyper-tuning in preventing overfitting or underfitting. It can be accomplished by using grid search or random search techniques to identify the ideal set of hyperparameters that will yield the best decision tree model performance.

```
Best hyperparameters: {'criterion': 'gini', 'max_depth': None, 'max_features': None, 'min_samples_leaf': 1, 'min_sa
mples_split': 2}
```

- **Gradient Boosting:**

Various hyperparameters have the potential to affect how well the gradient-boosting model performs. There are various parameters, including the total number of weak learners (trees), the learning rate (shrinkage), the maximum tree depth, and other factors. The learning rate, which governs the amount that each weak learner contributes to the ensemble, is a key hyperparameter.

It is important to balance model complexity and generalisation when choosing the best set of hyperparameters for Gradient Boosting. Grid search or random search techniques can be used to identify an ideal set of hyperparameters.

```
Best hyperparameters: {'max_depth': None, 'n_estimators': 150}
```

After the hyperparameter tuning was performed, the models were trained using these parameters and the final results were calculated using metrics such as Accuracy, F1 -score, Precision, Recall and AUC -ROC score. After determining which customers were more likely to leave, the next stage in the process was to determine the Customer Lifetime Value.

3. Calculating the Customer Lifetime Value (CLV):

Customer Lifetime Value, or CLV, is a crucial indicator that determines the overall monetary value that a customer is anticipated to produce for a company over a period of time.

The process of determining customer lifetime value is time-consuming. The monthly revenue for each churned customer is first calculated by dividing the total revenue by the duration of their tenure in months.

$$Monthly\ Revenue = \frac{Total\ Revenue}{Tenure\ in\ Months} \tag{13}$$

The churn rate was then determined by dividing the number of churned customers by the total number of customers.

$$Churn\ Rate = \frac{Number\ of\ Churned\ Customers}{Total\ number\ of\ Customers} \tag{14}$$

The reciprocal of the churn rate is then used to calculate the average customer lifespan, which represents the typical length of time a customer is active before churning.

$$Average\ Customer\ Lifespan = \frac{1}{Churn\ Rate} \tag{15}$$

A discount rate is chosen to reflect the time value of money. Finally, the CLV is then determined for each churned customer using a formula that accounts for the monthly revenue, average lifetime, discount rate, and churn rate.

$$CLV = \frac{Monthly\ Revenue\ \times\ Average\ Customer\ Lifespan}{1 + Discount\ Rate - Churn\ Rate} \tag{16}$$

This estimate gives information about the potential financial value that each customer may produce over the course of their relationship with the company.

VII. MODEL DEVELOPMENT AND EVALUATION

The models were further developed utilising the best parameters as explained above after the hyperparameters were obtained using the grid search method. The models were also assessed using metrics including accuracy, F1-score, precision, recall, and the AUC-ROC score. Below is a more detailed explanation of these measures used to assess the model's performance. Here, the letters TP, TN, FP, and FN stand for True Positives, True Negatives, False Positives, and False Negatives, respectively.

- **Accuracy:**

The most common classification metric used is accuracy. The number of accurate predictions made by the model to the total number of predictions made is the accuracy of the model. For instance, the accuracy of the model is 90/100 = 0.9 or 90% if it accurately predicts 90 out of 100 samples in a test dataset. Although accuracy is an easy-to-understand metric, it may not always be the most accurate indicator of a model's performance, particularly when the dataset is unbalanced. In these circumstances, different metrics like precision, recall, F1-score, or AUC -ROC score may be more appropriate.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

- **F1 – Score:**

The F1 score strikes an equilibrium between the trade-off between recall and precision, giving a comprehensive assessment of a model's performance. It's calculated as the harmonic mean of recall and precision.

$$F1\ Score = 2 \times\ (Precision\ \times\ Recall)/(Precision + Recall) \tag{18}$$

- **Precision:**

In binary classification issues, precision is a widely used evaluation metric that counts the proportion of accurately predicted positive instances among all positive predictions. In circumstances when false positives cost more than false negatives, precision is a relevant statistic.

$$Precision = \frac{TP}{TP + FP} \tag{19}$$

- **Recall:**

Recall is a metric that assesses a model's capacity to identify each relevant instance within a dataset. Other names for it include sensitivity and true positive rate. It can be expressed mathematically as the ratio of true positive (TP) cases to the total of both true positive and false negative (FN) cases.

$$Recall = \frac{TP}{TP + FN} \tag{20}$$

- **AUC – ROC (Area Under the Reciever Operating Characteristic Curve):**

AUC-ROC evaluates a model's capacity to distinguish between positive and negative classes across a range of thresholds. The relationship between the true positive rate and the false positive rate is plotted, and the area under the curve represents the model's discriminatory capacity. AUC-ROC value of 1.0 indicates the ideal classifier.

$$AUC\ ROC = \int_0^1 f(TP)\, d(FP) \tag{21}$$

The scores obtained by the models were as shown below,

Model	Accuracy	F1 -Score	Precision	Recall	AUC-ROC
Random Forest	97.47%	97.89%	98.90%	96.90%	97.62%
Decision Tree	95.89%	96.55%	98.35%	94.81%	96.19%
Gradient Boosting	95.59%	96.28%	98.28%	94.37%	95.92%
k-Nearest Neighbour	95.51%	96.20%	98.60%	93.92%	95.94%
Support Vector Machine	84.05%	86.71%	87.55%	85.89%	83.56%
Neural Network	77.27%	81.20%	81.38%	81.02%	84.66%

Table 1 Comparison of Model Performance Metrics.

As shown in the above table, the highest accuracy was obtained by the Random Forest model (97.47%). This was followed by the Decision Tree model (95.89%) with the Gradient Boosting and k-Nearest Neighbour models not far behind having accuracies of 95.59% and 95.51% respectively. The Support Vector Machine model and Neural Network model show lower accuracies in comparison, 84.05% and 77.27% respectively. Checking the F1 score, Precision, and recall of the top four models namely Random Forest (F1 score: 97.89%, Precision: 98.90%, Recall: 96.90%), Decision Tree (F1 score: 96.55%, Precision: 98.35%, Recall: 94.81%), Gradient Boosting (F1 score: 96.28%, Precision: 98.28%, Recall: 94.37%) and k – Nearest Neighbour (F1 score: 96.20%, Precision: 98.60%, Recall: 93.92%) it was found that the top performing models among these were the first three. Hence, from the above metrics it is concluded that Random Forest, Decision Tree and Gradient Boosting models worked the best to classify between the churned and unchurned customers. The below figure shows the metric scores of all the models,

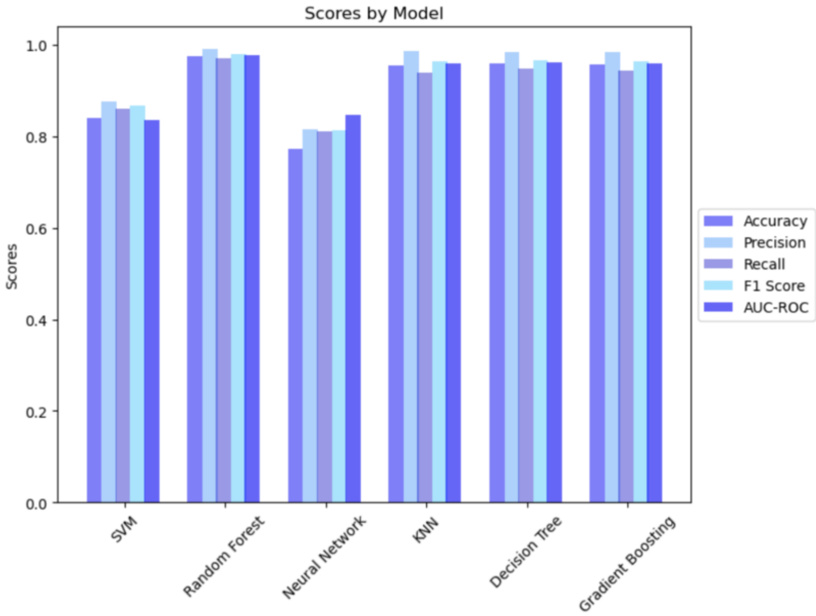


Figure 26 Metrics Scores Across Different Models.

The models were further assessed by then plotting their AUC – ROC scores as shown below to show their diagnostic ability as binary classifiers.

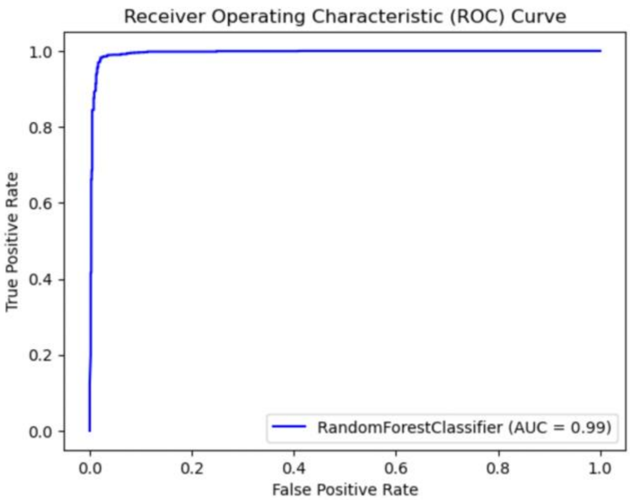


Figure 27 ROC for Random Forest Classifier

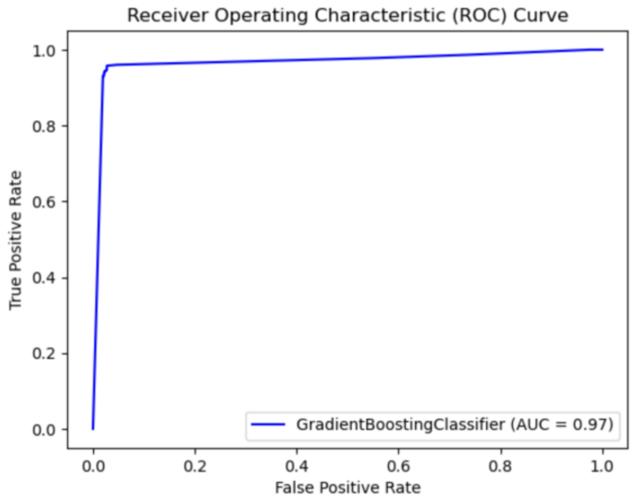


Figure 28 ROC for Gradient Boosting Classifier

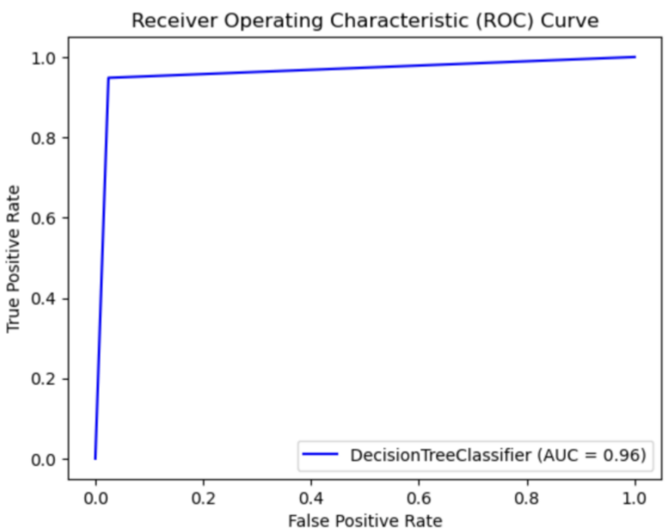


Figure 29 ROC for Decision Tree Classifier

The performance of the classification models is demonstrated by the AUC (Area Under Curve). Hence, from the results, we can conclude that the Random Forest classifier, which has an AUC value of 0.99, performs the best among the three classifiers. With AUC scores of 0.97 and 0.96, respectively, the Gradient Boosting Classifier and Decision Tree Classifier do not perform as well as the Random Forest classifier. Lastly, the confusion matrix was employed to gauge how the classifiers predicted the True positives, True Negatives, False Positives and False Negatives as shown in the figures below,

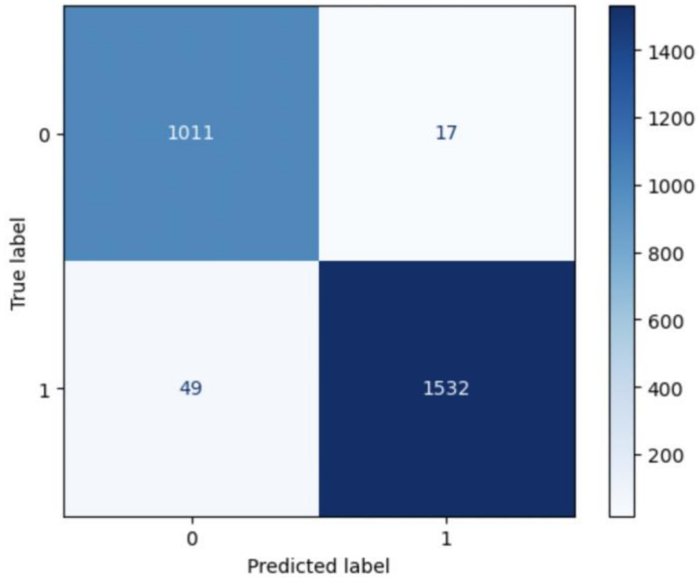


Figure 30 Confusion Matrix for Random Forest Classifier

The above confusion matrix shows the Random Forest Classifier has a high number of True Positives (1011) and True Negatives (1532). It has a very less rate of False Positives (49) and False Negatives (17).

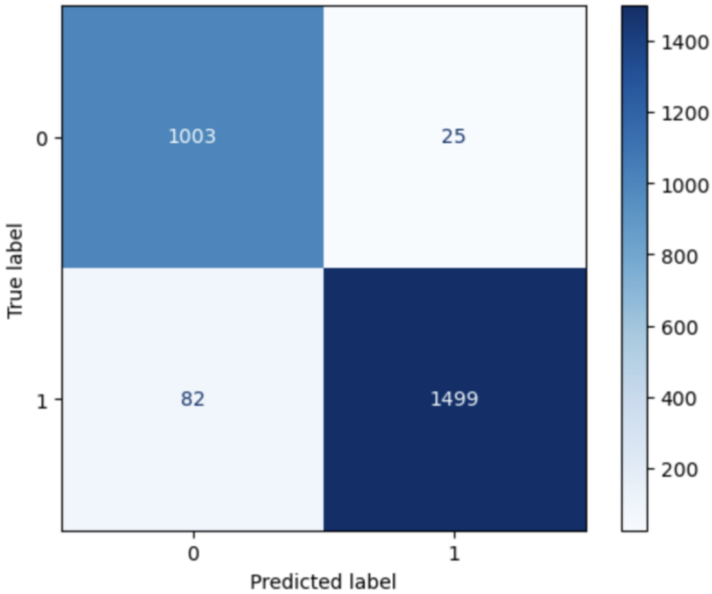


Figure 31 Confusion Matrix for Decision Tree Classifier.

The Decision Tree Classifier also has a high number of True Positives (1003) and True Negatives (1499) like the Random Forest Classifier. However, it has a higher number of False Positives (82) and False Negatives (25) as compared to the Random Forest classifier.

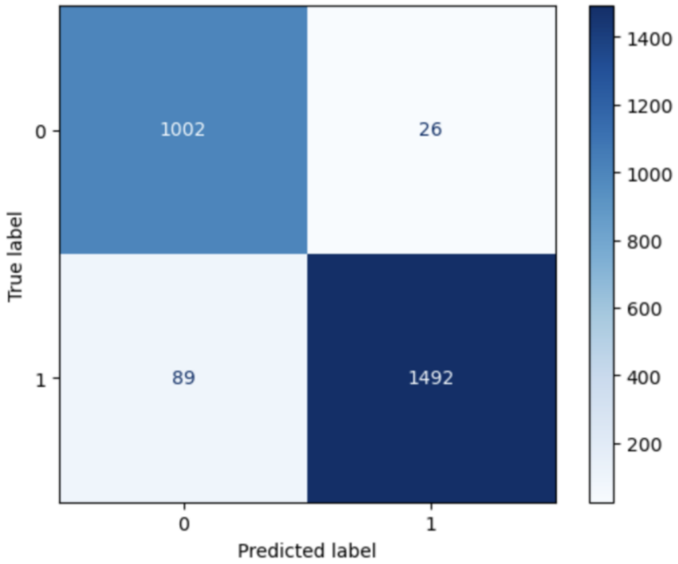


Figure 32 Confusion Matrix for Gradient Boosting Classifier.

The Gradient Boosting Classifier has a marginally low number of True Positives(1002) and True Negatives (1492) when compared to the Decision Tree Classifier. It has the highest number of False Positives (89) and False Negatives (26) as compared to the previous classifiers.
From these matrices, we can conclude that the Random Forest Classifier performs exceptionally well when compared to the rest of the classifiers that were trained. Hence, the Random Forest Classifier was further used on the validation set which was prepared at the start of the model evaluation phase.

VIII. Results And Discussion

This section delves deeply into the results of the machine learning models, highlighting the key factors influencing customer churn and offering insights to aid in strategic decision-making. We need to conduct a comprehensive analysis of these results in order to develop successful retention tactics and to further the ultimate goal of improving company performance. We gain a deeper understanding of customer habits in the process.
The Random Forest was selected and used to predict the probability of customer churn on the validation dataset. The model provided promising results. The threshold was then set and A new column called 'churned' was then made with the results. Values >0.5 were addressed as churned (True) and stayed (False). Figure 33 provides an overview of the outcome of the model results.

	predictions	churned
2882	0.967591	True
2602	0.053810	False
2614	0.013333	False
1981	0.088386	False
16208	0.994883	True
14730	0.964563	True
2518	0.805336	True
17379	0.997924	True
2974	0.026667	False
1713	0.026667	False
13651	0.123126	False
326	0.013333	False
6292	0.249737	False
14627	0.984566	True
1302	0.000000	False
12567	0.026667	False
8910	0.053333	False
9487	0.381285	False
3449	0.006667	False
4721	0.429454	False

Figure 33 Churn Probability Prediction Results

Figure 34 shows the resulting data frame for ‘df_val’ after resetting the index.

	index	Customer ID	predictions	churned
0	2882	9777-IQHWP	0.967591	True
1	2602	6035-BXTTY	0.053810	False
2	2614	2220-IAHLS	0.013333	False
3	1981	3675-EQOZA	0.088386	False
4	16208	4822-RVYBB	0.994883	True
5	14730	9565-DJPIB	0.964563	True
6	2518	7130-YXBRO	0.805336	True
7	17379	8375-DKEBR	0.997924	True
8	2974	8008-OTEZX	0.026667	False
9	1713	3458-IDMFK	0.026667	False

Figure 34 Dataframe after resetting index values.

This data frame was then merged with the original data frame on the ‘Customer ID’ column. A new data frame called ‘Churned’ was then created by setting the threshold to 0.5 which consisted of only the customers who showed a probability to churn. The Customer Lifetime Value Analysis was then performed on the resulting ‘Churned’ data frame. The CLV is a pivotal metric to understand the potential value of retaining these customers.
The Percentage of Revenue contribution by the top contributing customers was then calculated from the ‘Churned’ data frame. The top customers contributed 0.4% of the overall revenue which is a significant portion Figure 35 illustrates the results.

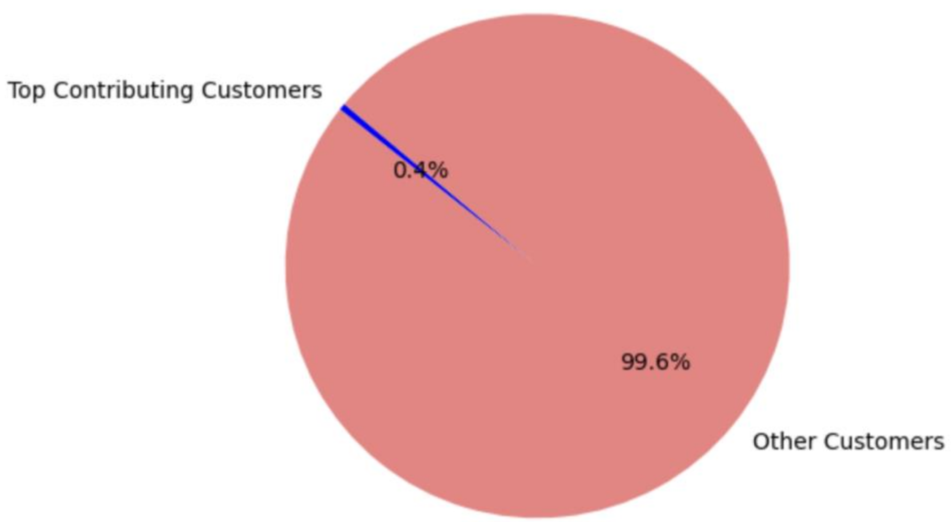


Figure 35 Percentage of Revenue Contribution by Top Contributing Customers.

To provide a more in-depth perspective, the CLV value was then used to calculate how much the top 5 customers were contributing to the total revenue. Their cumulative revenue contribution was about 2.3%. Figure 36 shows the results of this,

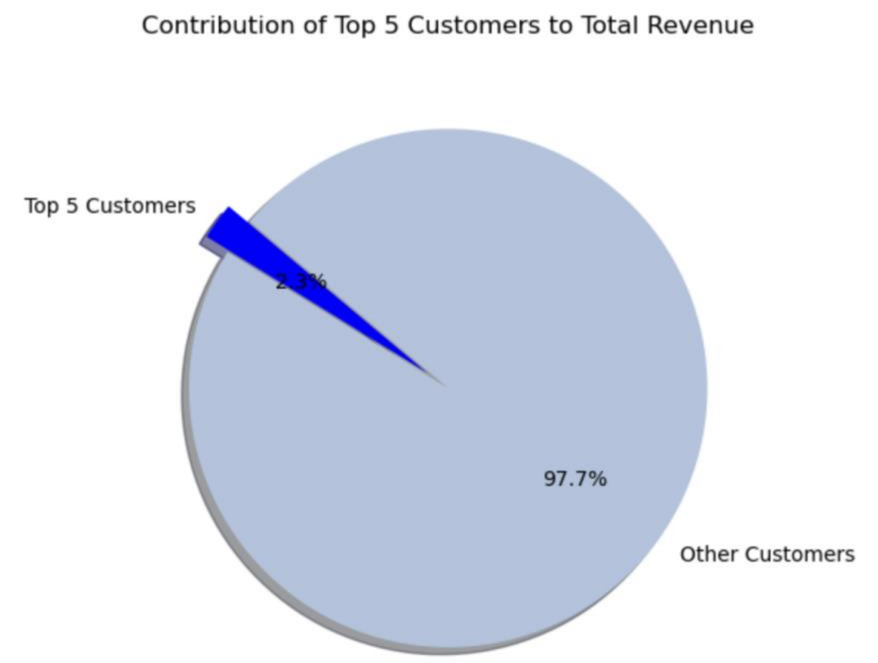


Figure 36 Contribution of Top 5 Customers with the Highest CLV.

Visualizing the results, a pie chart was made presenting the percentage of revenue contribution by the Top 5 customers along with their Customer ID as shown in Figure 37.

	Customer ID	CLV	churned
466	2239-JALAW	432.419943	True
2548	0330-IVZHA	274.224957	True
5596	9747-DDZOS	272.134323	True
2662	0491-KAPQG	223.376489	True
5588	2187-PKZAY	219.883979	True

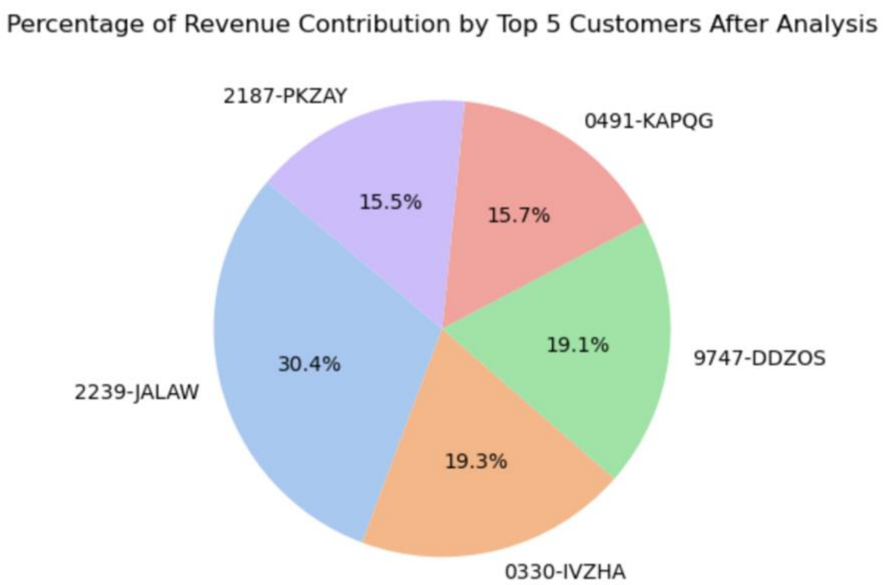


Figure 37 Percentage of Revenue Contribution by Top 5 Customers After Analysis.

By donating a significant 33.2% of the total 0.3% that is attributed to the top contributors, Customer ID '0078-XZMHT,' who is the most important contributor, emerges as the top contributor. On the other hand, '7426-WEIJX' plays a relatively minor part, contributing 8.7% to this total amount.

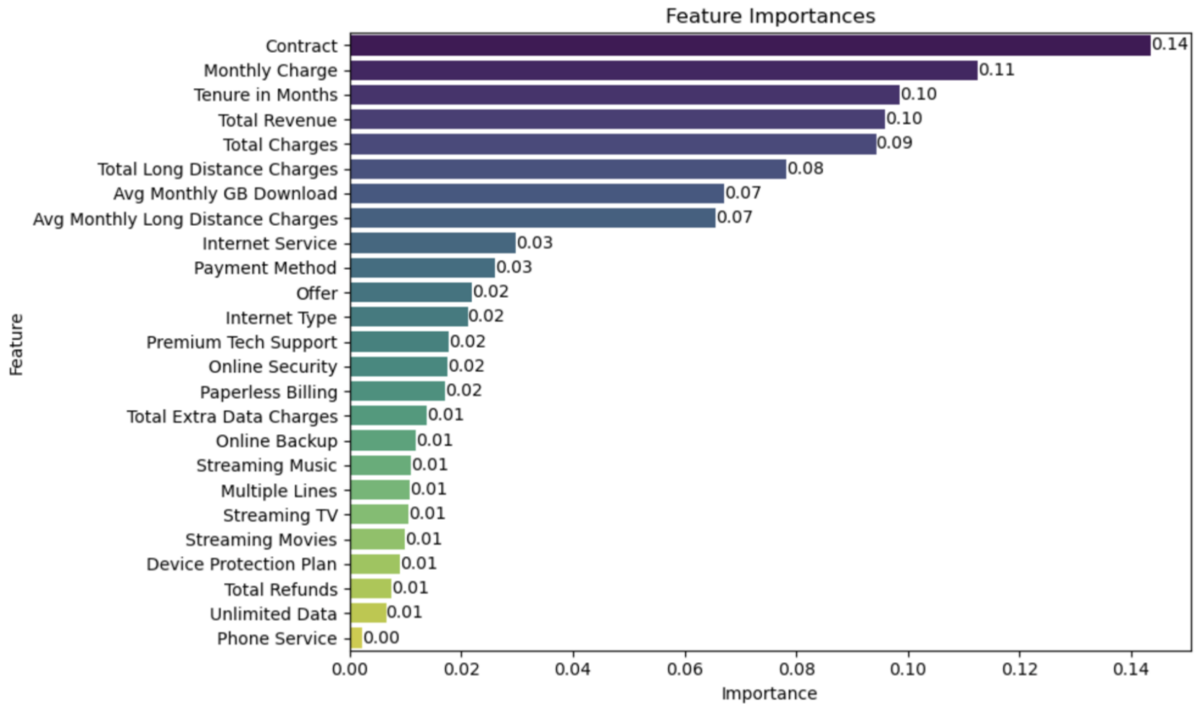


Figure 38 Feature Importance based on the Random Forest Classifier Implemented.

The Random Forest Classifiers feature importance scores highlight the significant factors that affect customer churn (Figure 38). Notably, the attributes like Contract, Monthly Charges, Tenure in Months, Total Revenue, Total Charges and Total Long Distance Charges have a substantial effect on churn. Meaning, Longer contracts and greater customer spending show strong correlations with lower churn, reflecting the loyalty and engagement of the customers. Moreover, the types of services preferred, payment methods and certain offers also moderately contribute to the prediction. On the other hand, phone service displays minimal impact to churn. Therefore, the business can profit by having targeted interventions that can involve offering attractive contract plans, optimizing the pricing strategies, and enhancing the quality of service to yield higher revenue and long-standing customers. Additionally, strengthening security features, streamlining communication channels, and implementing value-added services could also help reduce churn.

	Customer ID	Customer Status
95	8745-PVESG	1
7138	0164-APGRB	1
5879	8608-OZTLB	1
12922	8263-QMNTJ	1
3879	0786-IVLAW	1

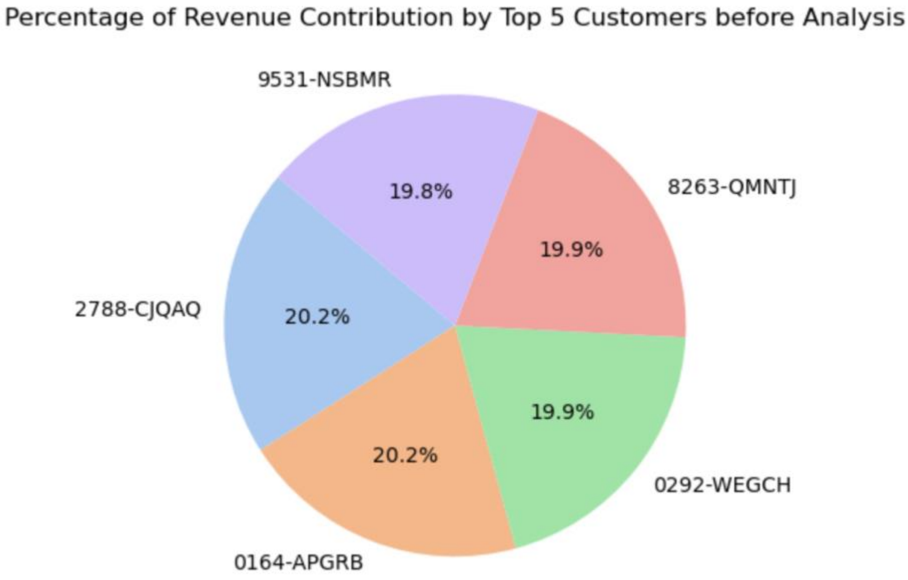


Figure 39 Percentage of Revenue Contribution by Top 5 Customers Before Analysis

An extra pie chart (Figure 39) showing the distribution of the top contributors in the primary dataset before the analysis was inserted in order to verify the analysis's robustness. It was clear that the top contributors found after the research did not profoundly overlap with the top contributors reported before the analysis.

	Customer ID	CLV	Customer Status
6863	9840-EFJQB	466.567701	1
13906	9750-BOOHV	466.567701	1
4884	1459-QNFQT	442.494407	1
11927	6878-GGDWG	442.494407	1
5035	8715-KKTFG	439.238864	1

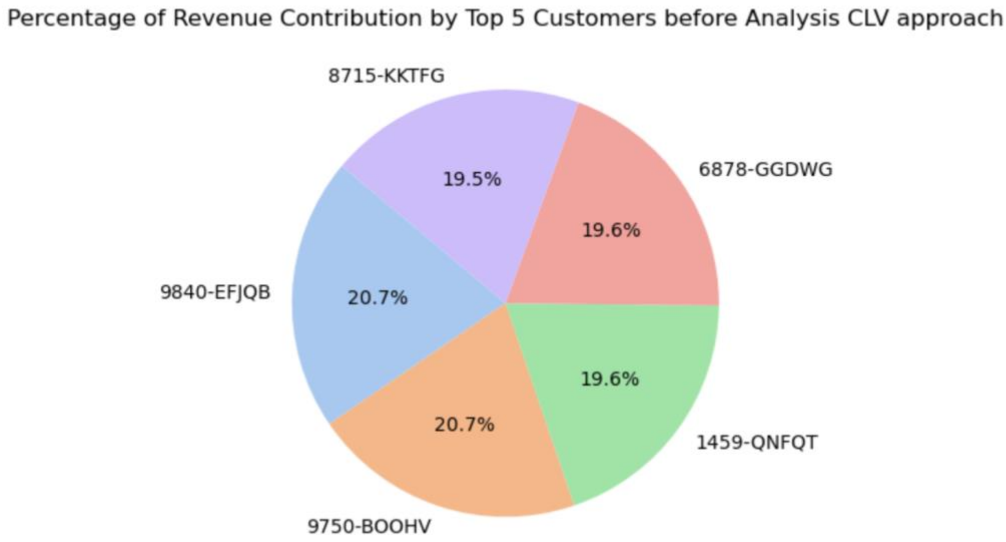


Figure 40 Percentage of Revenue Contribution by Top 5 Customers before Analysis using CLV Approach.

Again, the results were contrasted with a chart showing the top contributors only using the application of Customer Lifetime Value without utilising the Random Forest algorithm to emphasise the success of the employed approach (Figure 40). The top customers identified were clearly different across the two methodologies, which strengthened the validity of the analysis.

The comprehensive analysis of customer churn prediction and subsequent CLV calculation not only highlighted the potential revenue impact but also the distinctiveness of the identified top contributors when incorporating advanced modelling techniques, illuminating the analytical value in practical applications.

IX. LIMITATIONS AND FUTURE WORK.

- **Limitations**

The importance of acknowledging a study's limitations cannot be overstated; this section gives a summary of those that were encountered. While the selected machine learning models offer insight into customer churn, they are limited by their reliance on past data and patterns and lack access to current trends in customer behaviour. The quality and completeness of the dataset have a direct impact on how accurate the models are. As a result, the model's ability to predict the churn accurately can be impacted by any missing or noisy data.

Secondly, a time series analysis could not be performed since the dataset lacked a defined time range. A time series analysis could reveal temporal trends in consumer behaviour. The dataset further lacked information on the causes of customer churn. Despite some data being provided, it had to be excluded due to the large amount of missing data in the column. This made it harder to identify the precise causes of the problem. The approach primarily emphasizes quantitative characteristics.

Finally, while the discovered correlations offer valuable insight, they do not suggest causation. The study does not take into account external factors such as market trends or competitors, which may potentially have an impact on customer churn in a business.

- **Future Work**

This study provides insightful information on predicting customer churn, however, there are other ways to predict the churn. First, the analysis can use more sophisticated machine learning and ensemble methodologies to improve prediction power and accuracy further. More robust models can also result from optimizing the feature engineering process by taking the correlations between the various variables into account and incorporating domain-specific insights.

Deeper insights into the dynamics of customer turnover can also be gained by examining fluctuating aspects of customer behaviour, such as examining churn patterns over the course of multiple quarters or seasons and conducting sentiment analysis. A/B testing could also empirically confirm the effectiveness of retention methods developed in context with these findings.

X. CONCLUSION

In conclusion, our investigation not only employed a variety of machine learning models to accurately estimate the likelihood of customer churn in the telecom industry, but also used Customer Lifetime Value (CLV) to identify the top revenue drivers. With a 97% accuracy rate, the Random Forest Classifier emerged as the most effective machine learning model for achieving this. It was possible to gain insightful information by combining probabilistic churn estimates with the CLV analysis, which highlighted clients like "2239-JALAW" as significant revenue generators.

This investigation has a great deal of potential for the telecom sector, given these conclusions. Through efficient resource allocation to high-value customers, the dual strategy improves decision-making. Furthermore, by using this strategy, focused marketing campaigns that concentrate on high-value customers can optimise their marketing expenses.

Even though the study reveals key insights, it is crucial to talk about its shortcomings. It is challenging to perform sentiment analysis and evaluate temporal patterns in the absence of time range data and churn reason. These data-driven initiatives have the ability to drive sustained revenue while also boosting customer satisfaction and loyalty in the telecom industry, which is plagued by fierce competition and evolving customer preferences.

XI. DASHBOARD CREATION

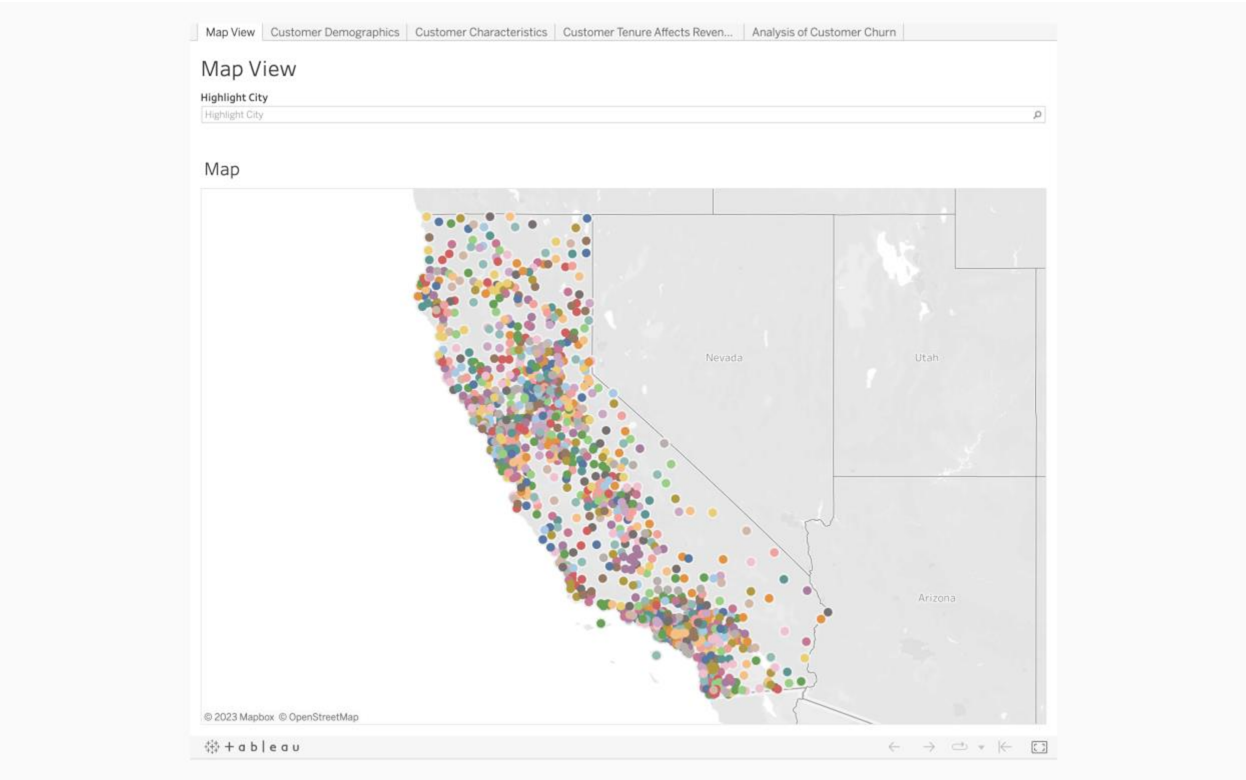
The creation of an informative dashboard is an important component of the analysis. It enables the presentation of complex results to stakeholders and decision-makers who may not be technically sound. The primary goal of the dashboard is to visually communicate the findings of the analysis and the insights derived from the study.

- **Data Sources**
  The dashboard is made from static sources, including the cleaned main dataset and the supplementary data derived from the analysis conducted in Jupyter Notebook. This ensures that the stakeholders are presented with up-to-date and relevant information.

- **Interactivity**
  The dashboard contains filters and highlighters to allow the free exploration of the data. Hovering over specific data points provides additional details, enhancing user engagement.

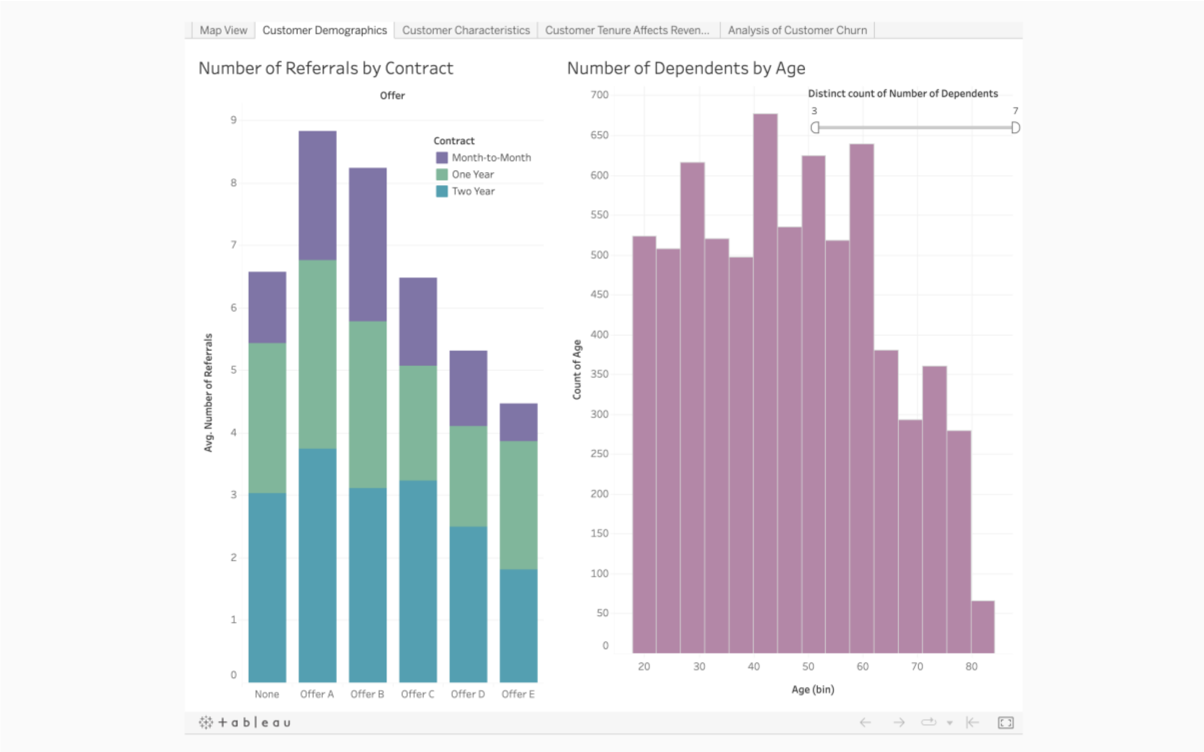
- **Dashboard Components**

i. Map View:
The first dashboard shows a geographical representation of the distribution of customers along with the count of customers residing at the respective Zip code.



Dashboard 1. Map View

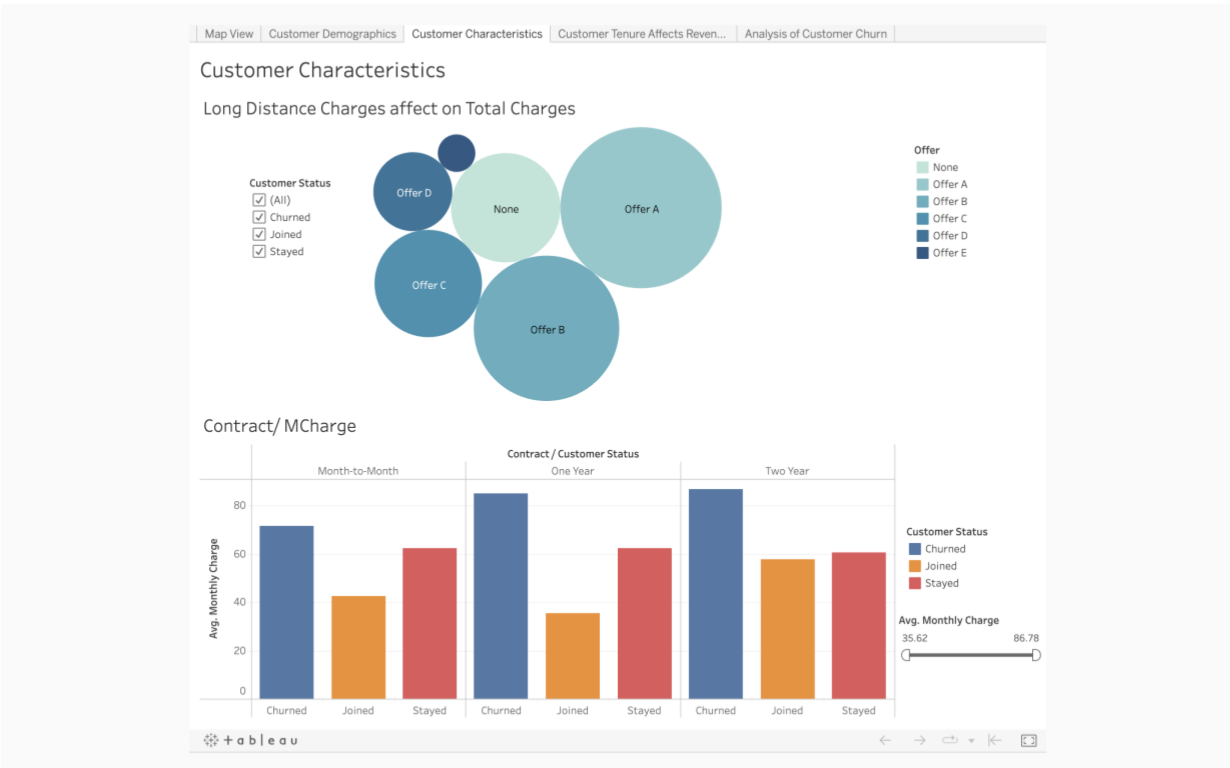
ii. Customer Demographics:
The second dashboard shows the customer demographics presenting insights using stacked bar graphs and bar graphs showing the "Number of Referrals by Contract" and the "Number of Dependents by Age".



Dashboard 2. Customer Demographics

iii. Customer Characteristics:

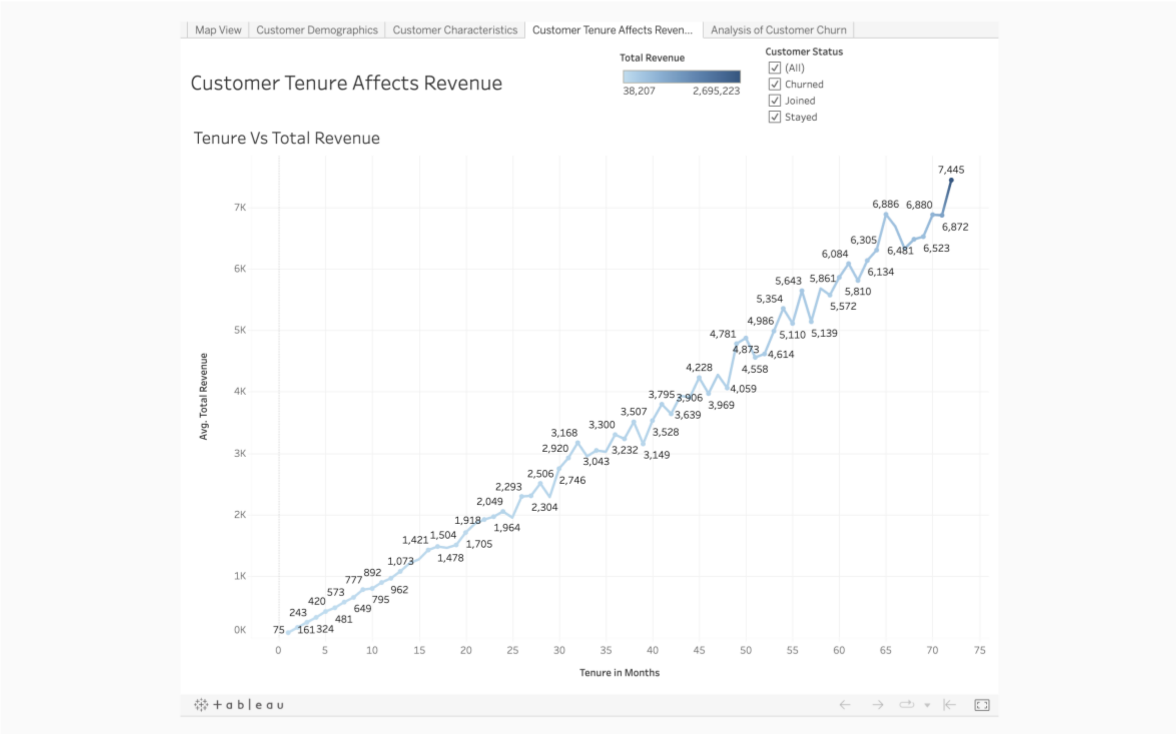
The third dashboard explores the customer characteristics showing the impact of "Long distance Charges" on the customer's overall Total Charges. It also displays different contracts and how they affect the monthly charges. It also explores the relationship between the monthly charge and customer status.



Dashboard 3. Customer Characteristics

iv. Customer Tenure Analysis:

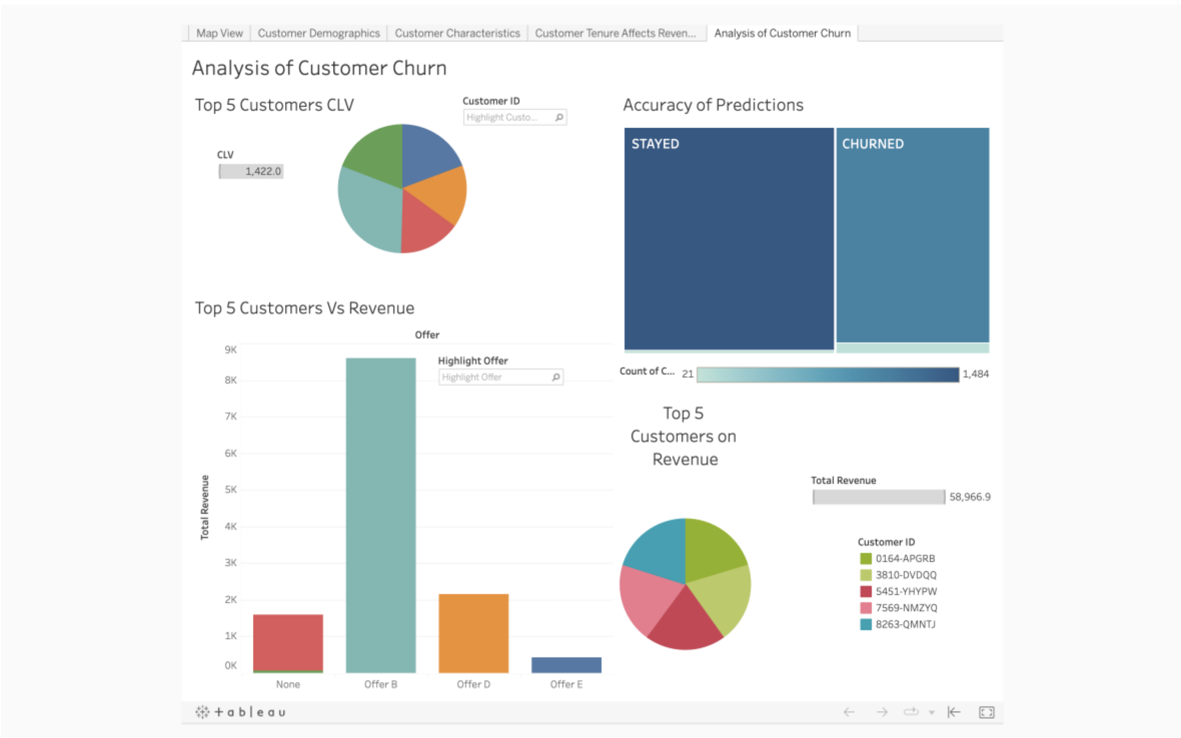
The fourth dashboard focuses on the Tenure of the Customers and the Total Revenue. This is represented by a line graph.



Dashboard 4. Customer Tenure Analysis

v. Customer Churn Analysis:

The fifth dashboard provides a comprehensive analysis of customer churn. The findings are represented with pie charts and bar graphs. The accuracy of the prediction model is shown in the form of a confusion matrix.



Dashboard 5. Customer Churn Analysis

XII. REFERENCES

Al-Mashraie, M., Chung, S. H., & Jeon, H. W. (2020). Customer switching behaviour analysis in the telecommunication industry via push-pull-mooring framework: A machine learning approach. Computers & Industrial Engineering, 144, 106476.

Bhuse, P., Gandhi, A., Meswani, P., Muni, R., & Katre, N. (2020, December). Machine learning based telecom-customer churn prediction. In 2020 3rd International Conference on Intelligent Sustainable Systems (ICISS) (pp. 1297-1301). IEEE.

Chaudhary, A., Rizvi, A., Kumar, N., & Mishra, A. K. (2023). A Novel Approach for Customer Churn Prediction in Telecom using Machine Learning Models.

Fujo, S. W., Subramanian, S., & Khder, M. A. (2022). Customer churn prediction in the telecommunication industry using deep learning. Information Sciences Letters, 11(1), 24.

Hawkins, D. L., & Hoon, S. (2019). The impact of customer retention strategies and the survival of small service-based businesses. Available at SSRN 3445173.

Jain, H., Khunteta, A., & Srivastava, S. (2020). Churn prediction in telecommunication using logistic regression and logit boost. Procedia Computer Science, 167, 101-112.

Khalid, L. F., Abdulazeez, A. M., Zeebaree, D. Q., Ahmed, F. Y., & Zebari, D. A. (2021, July). Customer churn prediction in telecommunications industry based on data mining. In 2021 IEEE Symposium on Industrial Electronics & Applications (ISIEA) (pp. 1-6). IEEE.

Kayaalp, F. (2017). Review of Customer Churn Analysis Studies in Telecommunications Industry. Karaelmas Science & Engineering Journal, 7(2).

Li, W., & Zhou, C. (2020, March). Customer churn prediction in telecom using big data analytics. In IOP Conference Series: Materials Science and Engineering (Vol. 768, No. 5, p. 052070). IOP Publishing.

Mohamed, F. A., & Al-Khalifa, A. K. (2023, January). A Review of Machine Learning Methods For Predicting Churn in the Telecom Sector. In 2023 International Conference On Cyber Management And Engineering (CyMaEn) (pp. 164-170). IEEE.

Peng, J., Quan, J., & Zhang, S. (2013). Mobile phone customer retention strategies and Chinese e-commerce. Electronic Commerce Research and Applications, 12(5), 321-327.

Prabadevi, B., Shalini, R., & Kavitha, B. R. (2023). Customer churning analysis using machine learning algorithms. International Journal of Intelligent Networks.

Saheed, Y. K., & Hambali, M. A. (2021, October). Customer churn prediction in telecom sector with machine learning and information gain filter feature selection algorithms. In 2021 International Conference on Data Analytics for Business and Industry (ICDABI) (pp. 208-213). IEEE.

Saroha, R., & Diwan, S. P. (2020). Development of an empirical framework of customer loyalty in the mobile telecommunications sector. Journal of Strategic Marketing, 28(8), 659-680.

Schaeffer, S. E., & Sanchez, S. V. R. (2020). Forecasting client retention—A machine-learning approach. Journal of Retailing and Consumer Services, 52, 101918.

Shafei, I., & Tabaa, H. (2016). Factors affecting customer loyalty for mobile telecommunication industry. EuroMed Journal of Business, 11(3), 347-361.

Sharma, A., Gupta, D., Nayak, N., Singh, D., & Verma, A. (2022, April). Prediction of Customer Retention Rate Employing Machine Learning Techniques. In 2022 1st International Conference on Informatics (ICI) (pp. 103-107). IEEE.

Taskin, N. (2023). CUSTOMER CHURN PREDICTION MODEL IN TELECOMMUNICATION SECTOR USING MACHINELEARNING TECHNIQUE.

Ullah, I., Raza, B., Malik, A. K., Imran, M., Islam, S. U., & Kim, S. W. (2019). A churn prediction model using random forest: analysis of machine learning techniques for churn prediction and factor identification in telecom sector. IEEE access, 7, 60134-60149.

VLN, R. K., & Deeplakshmi, P. (2021, January). Dynamic churn prediction using machine learning algorithms-predict your customer through customer behaviour. In 2021 International Conference on Computer Communication and Informatics (ICCCI) (pp. 1-6). IEEE.

Zadoo, A., Jagtap, T., Khule, N., Kedari, A., & Khedkar, S. (2022, May). A review on churn prediction and customer segmentation using machine learning. In 2022 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COM-IT-CON) (Vol. 1, pp. 174-178). IEEE

XIII. APPENDIX

University of Huddersfield
School of Computing and Engineering

PROJECT ETHICAL REVIEW FORM

Applicable for all research, master and undergraduate projects

Project Title:	Churn Prediction and Lifetime Value Analysis: A Data-Driven Client Retention Strategy for Telecom Companies.
Student:	Liyan Mohammed Mansooruddin Shaikh
Course/Programme:	MSc Data Analytics
Department:	School of Computing and Engineering
Supervisor:	Dr. Riazul Islam
Project Start Date:	22nd May 2023

ETHICAL REVIEW CHECKLIST

	Yes	**No**
1. Are there problems with any participant's right to remain anonymous?	☐	☒
2. Could a conflict of interest arise between a collaborating partner or funding source and the potential outcomes of the research, e.g. due to the need for confidentiality?	☐	☒
3. Will financial inducements be offered?	☐	☒
4. Will deception of participants be necessary during the research?	☐	☒
5. Does the research involve experimentation on any of the following?		
(i) animals?	☐	☒
(ii) animal tissues?	☐	☒
(iii)human tissues (including blood, fluid, skin, cell lines)?	☐	☒
6. Does the research involve participants who may be particularly vulnerable, e.g. children or adults with severe learning disabilities?	☐	☒
7. Could the research induce psychological stress or anxiety for the participants beyond that encountered in normal life?	☐	☒
8. Is it likely that the research will put any of the following at risk:		
(i) living creatures?	☐	☒
(ii) stakeholders (disregarding health and safety, which is covered by Q9)?	☐	☒
(iii)the environment?	☐	☒

(iv) the economy? ☐ ☒

9. Having completed a health and safety risk assessment form and taken all reasonable practicable steps to minimize risk from the hazards identified, are the residual risks acceptable (Please attach a risk assessment form – available at the end of this document). ☒ ☐

STATEMENT OF ETHICAL ISSUES AND ACTIONS

If the answer to any of the questions above is yes, or there are any other ethical issues that arise that are not covered by the checklist, then please give a summary of the ethical issues and the action that will be taken to address these in the box below. If you believe there to be no ethical issues, please enter "NONE".

Data privacy:
Concerns concerning data privacy arise when working with consumer information. To address this issue only necessary data will be collected for the project with the consent of the customer.

Data Security:
Throughout the project, precautions will be taken to ensure the data's security and confidentiality. Only authorized personnel will have access to sensitive information.

Responsible use of results:
The project's insights and predictions will be used in an ethical and responsible manner. It is important to take precautions to prevent manipulative or destructive uses of the forecasts.

STATEMENT BY THE STUDENT

I believe that the information I have given in this form on ethical issues is correct.

Signature: ______________________ Date: 17th May 2023

AFFIRMATION BY THE SUPERVISOR

I have read this Ethical Review Checklist and I can confirm that, to the best of my understanding, the information presented by the student is correct and appropriate to allow an informed judgement on whether further ethical approval is required.

Signature: Riazul Islam Date: 17/05/2023

SUPERVISOR RECOMMENDATION ON THE PROJECT'S ETHICAL STATUS

Having satisfied myself of the accuracy of the project ethical statement, I believe that the appropriate action is:

The project proceeds in its present form	✓
The project proposal needs further assessment by an Ethical Review Panel. The Supervisor will pass the form to the Ethical Review Panel Leader for consideration.	

RETENTION OF THIS FORM

- The Supervisor must retain a copy of this form until the project report/dissertation is produced.
- The student must include a copy of the form as an appendix in the report/dissertation.

OUTCOME OF THE ETHICAL REVIEW PANEL PROCESS, <u>WHERE REQUIRED</u>

Tick One

1. Approved. The ethical issues have been adequately addressed and the project may commence. ☐
2. Approved subject to minor amendments. The required amendments are stated in the box below. The project may proceed once the form has been amended in line with the requirements and signed by the Supervisor in the box imediately below to confirm this. ☐

I confirm, as Supervisor, that the amendments required have been made:

Signature: ______________________________________ Date: ____________________

3. Resubmit. The areas requiring further action are stated in the box below. The project may not proceed until the form has been resubmitted and approved. ☐
4. Reject. The reasons why it will not be possible to address the ethical issues adequately are stated in the box below. ☐

For any of the outcomes 2, 3 or 4 above, please provide a statement in the box below.

AFFIRMATION BY THE REVIEW PANEL LEADER

I approve the decision reached above by the review panel members:

Signature: ______________________________ Date: ____________________